

CSE 3203

Compiler Design and Automata Theory

Finite Automata, Language and Computation

- Complexity : What makes some problems computationally hard and others easy?
 - Answer: **We do not know**
 - But we have an elegant scheme to classify them
- Computability: Are there problems that are unsolvable
 - Answer: **Yes**
 - We also developed ideas concerning mathematical models of computers -> lead to actual development of computers
- Automata/Language Theory: Formally defining mathematical models of computation

In summary

Automata Theory → Computability → Complexity
(What is a computer?) (What can it do?) (How efficiently?)

Review

- Logic
- Sets
- Relations
- Functions
- Graphs
- Proof Techniques

Logic

- Logic defines a **formal language for representing knowledge** and for making logical inference
- It helps us to understand **how to construct a valid argument**

Logic defines:

- Syntax of statements
- The meaning of statements
- The rules of logical inference (manipulation)

Propositional logic

The simplest logic

- **Definition:**
 - A proposition is a statement that is either true or false.
- **Examples:**
 - JUST CSE department is on the first floor
 - **(T)**
 - $5 + 2 = 8$.
 - **(F)**
 - It is raining today.
 - **(either T or F)**

Propositional logic

Examples (cont.):

– How are you?

- a question is not a proposition

– $x + 5 = 3$

- since x is not specified, neither true nor false

– 2 is a prime number.

- (T)

– She is very talented.

- since she is not specified, neither true nor false

– There are other life forms on other planets in the universe.

- either T or F

Composite Statements

- More complex propositional statements can be built from elementary statements using **logical connectives**.

Example:

- Proposition A: **It rains outside**
- Proposition B: **We will see a movie**
- A new (combined) proposition:
If it rains outside then we will see a movie

Composite Statements

- More complex propositional statements can be built from elementary statements using **logical connectives**.
- **Logical connectives:**
 - Negation
 - Conjunction
 - Disjunction
 - Exclusive or
 - Implication
 - Biconditional

Logical connectives: Negation

Definition: Let **p** be a proposition. The statement "**It is not the case that p.**" is another proposition, called the **negation of p**. The **negation of p** is denoted by $\neg p$ and read as "**not p.**"

Example:

- JUST CSE department is on the first floor



- **It is not the case that** JUST CSE department is on the first floor.

Other examples:

- $5 + 2 \neq 8$.
- 10 is **not** a prime number.
- It is **not the case that** buses stop running at 9:00pm.

Logical connectives: Negation

A **truth table** displays the **relationships between truth values** (T or F) of different propositions.

P	$\neg p$
T	F
F	T

Rows: all possible values
of elementary proposition

Logical connectives: Conjunction

Definition: Let p and q be propositions. The proposition “ p and q ” denoted by $p \wedge q$, is true when **both p and q are true** and is false otherwise. The proposition $p \wedge q$ is called the **conjunction of p and q** .

- **Examples:**

- It is raining today **and** 2 is a prime number.
- 2 is a prime number **and** $5 + 2 \neq 8$.
- 13 is a perfect square **and** 9 is a prime.

Logical connectives: Disjunction

Definition: Let p and q be propositions. The proposition “ p or q ” denoted by $p \vee q$, is false when **both p and q are false** and is true otherwise. The proposition $p \vee q$ is called the **disjunction of p or q** .

- **Examples:**

- It is raining today **or** 2 is a prime number.
- 2 is a prime number **or** $5 + 2 \neq 8$.
- 13 is a perfect square **or** 9 is a prime.

Truth Tables

- **Conjunction and disjunction**
 - Four different combinations of values for p and q

p	q	$p \wedge q$	$p \vee q$
F	F		
F	T		
T	F		
T	T		

Rows: all possible combinations of values for elementary propositions: 2^n values

Truth Tables

- **Conjunction and disjunction**
 - Four different combinations of values for p and q

p	q	$p \wedge q$	$p \vee q$
F	F	F	
F	T	F	
T	F	F	
T	T	T	

Truth Tables

- **Conjunction and disjunction**
 - Four different combinations of values for p and q

p	q	$p \wedge q$	$p \vee q$
F	F	F	F
F	T	F	T
T	F	F	T
T	T	T	T

NB: $p \vee q$ (the **or** is used **inclusively**, i.e., $p \vee q$ is true when either p or q or both are true).

Exclusive or

Definition: Let p and q be propositions. The proposition " p exclusive or q " denoted by $p \oplus q$, is true when exactly one of p and q is true and it is false otherwise.

p	q	$p \oplus q$
F	F	F
F	T	T
T	F	T
T	T	F

Implications

Definition: Let p and q be propositions. The proposition " p implies q " denoted by $p \rightarrow q$ is called implication. It is **false** when p is true and q is false and is true otherwise.

In $p \rightarrow q$, p is called the **hypothesis** and q is called the **conclusion**.

p	q	$p \rightarrow q$
F	F	T
F	T	T
T	F	F
T	T	T

Implications

Logic wants implication to **fail only when a true hypothesis leads to a false conclusion.**

If the hypothesis is false, the implication does **not make any promise**, so it's automatically considered **true** (this is called ***vacuous truth***).

Implications

$p \rightarrow q$ is read in a variety of equivalent ways:

- if p then q or *if p , q*
- p only if q
- p is sufficient for q or *a sufficient condition for q is p*
- q whenever p or *q when p*
- *p implies q*
 - *q is necessary for p or a necessary condition for p is q*

Examples:

– if Germany won 2010 world cup then 2 is a prime.

- If F then T ? **T**

– if today is Monday then $2 * 3 = 8$.

- $T \rightarrow F$? **F**

Necessary and Sufficient

- A necessary condition for getting an A in 3203 is that a student must do the assignments
- A sufficient condition for getting an A in 3203 is that a student must do the assignments and attend classes and take all exams and work hard
- So what is the difference?
 - If the student gets an A that means he/she did the assignments
 - But just doing the assignments does not guarantee an A
 - But if the sufficient condition(s) is satisfied then an A is guaranteed

Necessary and Sufficient

- **Sufficient** → “Enough to guarantee the outcome” → if this happens, the result must happen.
- **Necessary** → “Must be there for the outcome to happen” → without it, the result is impossible.
But with it, is the outcome guaranteed?

Necessary and Sufficient

- If p is necessary for q then q cannot be true unless p is true
 - i.e., q is true only if p is true or $q \rightarrow p$ (only if condition)
- If p is sufficient for q then p's being true is enough to make q true
 - i.e., if p is true then q is true or $p \rightarrow q$ (if condition)
- Can something be a sufficient condition but not a necessary condition for some other thing?
 - i.e., p is sufficient for q but not necessary for q
 - Yes:
 - *Boiling potatoes in water is a sufficient condition for cooking them*
 - *But not a necessary condition: there can be other ways to cook them frying them, baking them*

Necessary and Sufficient

- Can something be a both necessary and sufficient condition for some other thing?
- Yes
 - Answering all the questions correctly on a test is both a necessary and sufficient condition for a perfect score on the test
 - Answering all questions correctly is a **necessary** condition to get a perfect score
 - Because you will not get a perfect score on the test unless you do not answer all questions correctly
 - Answering all questions correctly is a **sufficient** condition to get a perfect score
 - Because getting all the answers correct is enough to get a perfect score

Biconditional

Definition: Let p and q be propositions. The biconditional $p \leftrightarrow q$ (read p if and only if q), is true when p and q have the same truth values and is false otherwise.

p	q	$p \leftrightarrow q$
F	F	T
F	T	F
T	F	F
T	T	T

Note: two truth values always agree.

Biconditional

- So, $p \leftrightarrow q$ means $p \rightarrow q$ and $q \rightarrow p$
- Or, $p \leftrightarrow q$ is equivalent to $p \rightarrow q \wedge q \rightarrow p$

Implications

The **converse** of $p \rightarrow q$ is $q \rightarrow p$.

- The **contrapositive** of $p \rightarrow q$ is $\neg q \rightarrow \neg p$
- The **inverse** of $p \rightarrow q$ is $\neg p \rightarrow \neg q$

Examples:

- If it jams, the traffic moves slowly.
- p : it jams
- q : traffic moves slowly.
- $p \rightarrow q$

Implications

The converse:

If the traffic moves slowly then it jams.

- $q \rightarrow p$

The contrapositive:

• If the traffic does not move slowly then it does not jams.

- $\neg q \rightarrow \neg p$

The inverse:

• If it does not jams the traffic moves quickly.

- $\neg p \rightarrow \neg q$

Constructing the truth table

Example: Construct a truth table for $(p \rightarrow q) \wedge (\neg p \leftrightarrow q)$

- Simpler if we decompose the sentence to elementary and intermediate propositions

p	q	$\neg p$	$p \rightarrow q$	$\neg p \leftrightarrow q$	$(p \rightarrow q) \wedge (\neg p \leftrightarrow q)$
F	F				
F	T				
T	F				
T	T				

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p	q	$\neg p$	$p \rightarrow q$	$\neg p \leftrightarrow q$	$(p \rightarrow q) \wedge (\neg p \leftrightarrow q)$
F	F	T			
F	T	T			
T	F	F			
T	T	F			

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p	q	$\neg p$	$p \rightarrow q$	$\neg p \leftrightarrow q$	$(p \rightarrow q) \wedge (\neg p \leftrightarrow q)$
F	F	T	T		
F	T	T	T		
T	F	F	F		
T	T	F	T		

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p	q	$\neg p$	$p \rightarrow q$	$\neg p \leftrightarrow q$	$(p \rightarrow q) \wedge (\neg p \leftrightarrow q)$
F	F	T	T	F	
F	T	T	T	T	
T	F	F	F	T	
T	T	F	T	F	

Constructing the truth table

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- Simpler if we decompose the sentence to elementary and intermediate propositions

p	q	$\neg p$	$p \rightarrow q$	$\neg p \leftrightarrow q$	$(p \rightarrow q) \wedge (\neg p \leftrightarrow q)$
F	F	T	T	F	F
F	T	T	T	T	T
T	F	F	F	T	F
T	T	F	T	F	F

Checking validity of logical equivalence

- We can use Truth Tables to check validity of logical equivalence
- For example, earlier we said
 $p \leftrightarrow q$ is equivalent to $p \rightarrow q \wedge q \rightarrow p$
- How can we check the validity of this statement?
 - Column labeled $p \leftrightarrow q$ should be same as the column labeled $p \rightarrow q \wedge q \rightarrow p$

Computer Representation of True and False

We need to encode two values True and False:

- Computers represents data and programs using 0s and 1s
- Logical truth values – True and False
- A bit is sufficient to represent two possible values:
 - 0 (False) or 1(True)
- A variable that takes on values 0 or 1 is called a Boolean variable.
- Definition:A bit string is a sequence of zero or more bits. The length of this string is the number of bits in the string.

Bitwise operation

- T and F replaced with 1 and 0

p	q	$p \wedge q$	$p \vee q$
0	0	0	0
0	1	0	1
1	0	0	1
1	1	1	1

p	$\neg p$
0	1
1	0

Bitwise operation

- T and F replaced with 1 and 0

1 0 1 1 0 0 1 1	1 0 1 1 0 0 1 1	1 0 1 1 0 0 1 1
∨ 0 1 0 0 1 0 0 1	∧ 0 1 0 0 1 0 0 1	⊕ 0 1 0 0 1 0 0 1
1 1 1 1 1 0 1 1	0 0 0 0 0 0 0 1	1 1 1 1 1 0 1 0

Applications of propositional logic

- **Translation of English sentences**
- **Inference and reasoning:**
 - new true propositions are inferred from existing ones
 - Used in **Artificial Intelligence**:
 - Builds programs that act intelligently
 - Programs often rely on symbolic manipulations
- **Design of logic circuit**

Translation

Assume a sentence:

If you are older than 13 or you are with your parents then you can attend a PG-13 movie.

Parse:

- If (you are older than 13 or you are with your parents) then (you can attend a PG-13 movie)

Atomic (elementary) propositions:

- A= you are older than 13
- B= you are with your parents
- C=you can attend a PG-13 movie

- Translation: $A \vee B \rightarrow C$

Application of Inference

Assume we know the following sentences are true:

If you are older than 13 or you are with your parents then you can attend a PG-13 movie. You are older than 13.

Translation:

- If (you are older than 13 or you are with your parents) then (you can attend a PG-13 movie) . (You are older than 13).
- A= you are older than 13
- B= you are with your parents
- C=you can attend a PG-13 movie
- $(A \vee B \rightarrow C), A$
- $(A \vee B \rightarrow C) \wedge A$ is true
- With the help of the logic we can infer the following statement (proposition):
- You can attend a PG-13 movie or C is True

Tautology and Contradiction

Some propositions are interesting since their values in the truth table are always the same

Definitions:

- A **compound proposition that is always true** for all possible truth values of the propositions is called a **tautology**.
- A **compound proposition that is always false** is called a **contradiction**.
- A proposition that is **neither a tautology nor contradiction** is called a **contingency**.

Example: $p \vee \neg p$ is a **tautology**.
 $p \wedge \neg p$ is a **contradiction**.

p	$\neg p$	$p \vee \neg p$	$p \wedge \neg p$
1	0	1	0
0	1	1	0

Equivalence

How do we determine that **two propositions** are equivalent?

Their truth values in the truth table are the same.

- Example: $p \rightarrow q$ is equivalent to $\neg q \rightarrow \neg p$ (contrapositive)

p	q	$p \rightarrow q$	$\neg q \rightarrow \neg p$
F	F	T	T
F	T	T	T
T	F	F	F
T	T	T	T

Equivalent statements are important for logical reasoning since they can be substituted and can help us to make a logical argument.

Logical Equivalence

Definition: The propositions p and q are called logically equivalent if $p \leftrightarrow q$ is a tautology (alternately, if they have the same truth table). The notation $p \Leftrightarrow q$ denotes p and q are logically equivalent.

p	q	$p \rightarrow q$	$\neg q \rightarrow \neg p$	$(p \rightarrow q) \Leftrightarrow (\neg q \rightarrow \neg p)$
F	F	T	T	T
F	T	T	T	T
T	F	F	F	T
T	T	T	T	T

Important Logical Equivalence

- **Double negation**

- $\neg(\neg p) \iff p$

- **Commutative**

- $p \vee q \iff q \vee p$

- $p \wedge q \iff q \wedge p$

- **Associative**

- $(p \vee q) \vee r \iff p \vee (q \vee r)$

- $(p \wedge q) \wedge r \iff p \wedge (q \wedge r)$

Important Logical Equivalence

- **Distributive**

- $p \vee (q \wedge r) \Leftrightarrow (p \vee q) \wedge (p \vee r)$
- $p \wedge (q \vee r) \Leftrightarrow (p \wedge q) \vee (p \wedge r)$

- **De Morgan**

- $\neg(p \vee q) \Leftrightarrow \neg p \wedge \neg q$
- $\neg(p \wedge q) \Leftrightarrow \neg p \vee \neg q$

- **Other useful equivalences**

- $p \vee \neg p \Leftrightarrow T$
- $p \wedge \neg p \Leftrightarrow F$
- $p \rightarrow q \Leftrightarrow (\neg p \vee q)$

Showing Logical Equivalence

Show $(p \wedge q) \rightarrow p$ is a tautology

In other words $((p \wedge q) \rightarrow p \Leftrightarrow T)$

$(p \wedge q) \rightarrow p$	$\Leftrightarrow \neg(p \wedge q) \vee p$	Useful
	$\Leftrightarrow [\neg p \vee \neg q] \vee p$	DeMorgan
	$\Leftrightarrow [\neg q \vee \neg p] \vee p$	Commutative
	$\Leftrightarrow \neg q \vee [\neg p \vee p]$	Associative
	$\Leftrightarrow \neg q \vee [T]$	Useful
	$\Leftrightarrow T$	Domination

You can also use truth table to show this

Limitation of Propositional Logic

Propositional logic: the world is described in terms of elementary propositions and their logical combinations

Elementary statements:

- Typically **refer to objects, their properties and relations.**
- But these are not **explicitly represented** in the propositional logic
- Rusafi is a NSU student

Rusafi	- object
A NSU student	- a property

Objects and properties are hidden in the statement, it is not possible to reason about them

Limitation of Propositional Logic

Statements that must be repeated for many objects

Example

If Ali is a CSE NSU graduate then Ali has passed CSE173

Translation

Ali is a CSE NSU graduate $\rightarrow$ Ali has passed CSE173

Solution: make **statements with variables**

If x is a CSE NSU graduate then x has passed CSE173

x a CSE NSU graduate $\rightarrow$ x has passed CSE173

Limitation of Propositional Logic

Statements that define the property of the group of objects

Example

Some of the CSE graduates graduate with honors.

Solution: make **statements with quantifiers**

- **Universal quantifier** –the property is satisfied by all members of the group
- **Existential quantifier** – at least one member of the group satisfy the property

Predicate Logic

Remedies the limitations of the propositional logic

- **Explicitly models objects and their properties**
- Allows to make statements with **variables and quantify** them

Basic building blocks of the predicate logic:

- **Constant** –models a specific object

Examples: “Ali”, “Khulna”, “7”

- **Variable** – represents object of specific type
(defined by the *universe of discourse*)

Examples: x, y

(universe of discourse can be people, students, numbers)

- **Predicate** - Represents properties of objects

Examples: **Red**(car23), **student**(x), **married**(John,Ann)

Assigning value to Predicate

Predicate - Represents properties of objects

A predicate $P(x)$ assigns a value true or false to each x depending on whether the property holds or not for x .

x is a prime number (universe of discourse is integers)

What are the truth values of:

- $P(2)$ T
- $P(3)$ T
- $P(4)$ F
- $P(5)$ T

All statements $P(2)$, $P(3)$, $P(4)$, $P(5)$ are propositions.

Is $P(x)$ a proposition? **No**. Many substitutions are possible.

Assigning value to Predicate

Predicates can have **more arguments which represent the relations between objects**

Example:

- Let $Q(x,y)$ denote ' $x+5 > y$ '
 - Is $Q(x,y)$ a proposition? **No!**
 - Is $Q(3,7)$ a proposition? **Yes**
 - What is the truth value of:
 - $Q(3,7)$ T
 - $Q(1,6)$ F
 - $Q(3,y)$ may be T or F. **not a proposition.**

Predicate is not proposition

Important:

- statement **$P(x)$ is not a proposition** since there are many objects it can be applied to

This is the same as in propositional logic ...

... But the difference is:

- predicate logic **allows us to explicitly manipulate** and substitute for the objects
- Predicate logic **permits quantified sentences** where variables are substituted for statements about the group of objects

Quantified Statements

Predicate logic lets us to make statements about groups of objects

- To do this we use special quantified expressions

Two types of quantified statements:

- **universal**

Example: ‘ all CSE DU graduates have to pass CSE 3203’

- the **statement is true for all** graduates

- **existential**

Example: ‘Some CSE DU students graduate with cgpa of 4.0.’

- the **statement is true for some** people

Universal Quantifier

Defn: The universal quantification of $P(x)$ is the proposition:
" *$P(x)$ is true for all values of x in the domain of discourse.*"

The notation $\forall x P(x)$ denotes the universal quantification of $P(x)$, and is expressed as *for every x , $P(x)$.*

Example:

- Let $P(x)$ denote $x > x - 1$.
- What is the truth value of $\forall x P(x)$?
- Assume the universe of discourse of x is all real numbers.
- Answer: Since every number x is greater than itself minus 1.
Therefore, $\forall x P(x)$ is true.

Universally Quantified Statement

Predicate logic lets us make statements about groups of objects

Universally quantified statement

- $\text{CSE-major}(x) \rightarrow \text{Student}(x)$
 - Translation: “if x is a CSE-major then x is a student”
 - **Proposition: no.**

- $\forall x \text{ CSE - major}(x) \rightarrow \text{Student}(x)$
 - Translation: “(For all people it holds that) if a person is a CSE-major then she is a student.”
 - **Proposition: yes.**

Existential Quantifier

Definition: The existential quantification of $P(x)$ is the proposition

"There exists an element in the domain (universe) of discourse such that $P(x)$ is true."

The notation: $\exists x P(x)$

Example 1:

- Let $T(x)$ denote $x > 5$ and x is from Real numbers.
- What is the truth value of $\exists x T(x)$?

• Answer:

- Since $10 > 5$ is true. Therefore, it is true that $\exists x T(x)$

Quantified statements

Statements about groups of objects

Example:

- $\text{CSE-DU-graduate}(x) \wedge \text{A-grade-student}(x)$
 - **Translation:** “**x is a CSE-DU-graduate and x is an A grade student**”
 - **Proposition: no.**
- $\exists x \text{ CSE - DU - graduate}(x) \wedge \text{A - grade - student}(x)$
 - **Translation:** “There is a person who is a CSE-DU-graduate and who is also an A grade student.”
 - **Proposition: ? yes**

Translation with Quantifier

Sentence:

- All DU students are smart.
- **Assume:** the domain of discourse of x are DU students
- Translation:
 - $\forall x \text{ smart}(x)$
- **Assume:** the universe of discourse are **students** (all students):
 - $\forall x \text{ at}(x, DU) \rightarrow \text{smart}(x)$
- **Assume:** the universe of discourse are **people**:
 - $\forall x \text{ student}(x) \wedge \text{at}(x, DU) \rightarrow \text{smart}(x)$

Translation with Quantifier

Sentence:

- Someone at DU is smart.
- **Assume:** the domain of discourse are all DU affiliates
- Translation:
 - $\exists x \text{ smart}(x)$
- **Assume:** the universe of discourse are people:
 - $$\exists x \text{ at}(x, DU) \wedge \text{smart}(x)$$

Translation with Quantifier

Assume two predicates $S(x)$ and $P(x)$

Universal statements typically tie with implications

- All $S(x)$ is $P(x)$
 - $\forall x (S(x) \rightarrow P(x))$
- No $S(x)$ is $P(x)$
 - $\forall x (S(x) \rightarrow \neg P(x))$

Existential statements typically tie with conjunctions

- Some $S(x)$ is $P(x)$
 - $\exists x (S(x) \wedge P(x))$
- Some $S(x)$ is not $P(x)$
 - $\exists x (S(x) \wedge \neg P(x))$

Nested Quantifier

More than one quantifier may be necessary to capture the meaning of a statement in the predicate logic.

Example:

- Every real number has its corresponding negative.
- Translation:
 - Assume:
 - a real number is denoted as x and its negative as y
 - A predicate $P(x,y)$ denotes: “ $x + y = 0$ ”
- Then we can write:

$$\forall x \exists y P(x, y)$$

Order of Quantifier

The order of nested quantifiers matters if quantifiers are of different type

Example:

- Assume $F(x,y)$ denotes “ x is a friend of y ”
- Then: $\forall x \exists y F(x, y)$
- Translates to: Everybody is a friend of somebody.
- And: $\exists y \forall x F(x, y)$
- Translates to: There is someone who is friend of everyone.

The meaning of the two is different.

Order of Quantifier

The order of nested quantifiers **does not matter** if quantifiers are of the same type

Example:

- For all x and y , if x is a parent of y then y is a child of x
- Assume:
 - $\text{Parent}(x,y)$ denotes “ x is a parent of y ”
 - $\text{Child}(x,y)$ denotes “ x is a child of y ”

- Two equivalent ways to represent the statement:

$$\forall x \forall y \text{Parent}(x, y) \rightarrow \text{child}(y, x)$$

$$\forall y \forall x \text{Parent}(x, y) \rightarrow \text{child}(y, x)$$

Negation of quantifiers

English statement:

- **Nothing is perfect.**
- Translation: $\neg \exists x \text{Perfect}(x)$

Another way to express the same meaning:

- **Everything is imperfect.**
- Translation: $\forall x \neg \text{Perfect}(x)$

Conclusion: $\neg \exists x P(x)$ is equivalent to $\forall x \neg P(x)$

Negation of quantifiers

DeMorgan Laws for quantifiers

Negation	Equivalent
$\neg \exists x P(x)$	$\forall x \neg P(x)$
$\neg \forall x P(x)$	$\exists x \neg P(x)$

Rules of Inference

Rules of inference: logically valid inference patterns

Modus Ponens, or the Law of Detachment

- Rules of inference

p

$p \rightarrow q$

Therefore, q

- Given p is true and the implication $p \rightarrow q$ is true then q is true.

Rules of Inference

Addition

$$p \rightarrow (p \vee q)$$

p

Therefore, $p \vee q$

Simplification

$$p \wedge q \rightarrow p$$

$p \wedge q$

Therefore, p

Rules of Inference

Modus Tollens

$$[\neg q \wedge (p \rightarrow q)] \rightarrow \neg p$$

Hypothetical Syllogism

$$[(p \rightarrow q) \wedge (q \rightarrow r)] \rightarrow (p \rightarrow r)$$

Disjunctive Syllogism

$$[(p \vee q) \wedge \neg p] \rightarrow q$$

Logical Equivalence

$$p \Leftrightarrow q$$

$P \rightarrow q$ is a tautology

DeMorgans law

$$\neg(p \vee q) \Leftrightarrow \neg p \wedge \neg q$$

Applying rule of Inference

Text:

- (1) It is not sunny this afternoon and it is colder than yesterday.
- (2) We will go swimming only if it is sunny.
- (3) If we do not go swimming then we will take a canoe trip.
- (4) If we take a canoe trip, then we will be home by sunset.

Propositions:

- 1.p = It is sunny this afternoon,
- 2.q = it is colder than yesterday,
- 3.r = We will go swimming ,
- 4.s= we will take a canoe trip
- 5. t= We will be home by sunset

Applying rule of Inference

Propositions:

p = It is sunny this afternoon,

q = it is colder than yesterday,

r = We will go swimming ,

s= we will take a canoe trip

t= We will be home by sunset

Translation:

- (1) $\neg p \wedge q$,
 - (2) $r \rightarrow p$,
 - (3) $\neg r \rightarrow s$,
 - (4) $s \rightarrow t$
- (1) It is not sunny this afternoon and it is colder than yesterday.
 - (2) We will go swimming only if it is sunny.
 - (3) If we do not go swimming then we will take a canoe trip.
 - (4) If we take a canoe trip, then we will be home by sunset.

We want to show: t

Applying rule of Inference

Proof:

1. $\neg p \wedge q$	Hypothesis
2. $\neg p$	Simplification
3. $r \rightarrow p$	Hypothesis
4. $\neg r$	Modus tollens (step 2 and 3)
5. $\neg r \rightarrow s$	Hypothesis
6. s	Modus ponens (steps 4 and 5)
7. $s \rightarrow t$	Hypothesis
8. t	Modus ponens (steps 6 and 7)
end of proof	

Set

Definition:

A set is a (unordered) collection of objects.

These objects are sometimes called **elements or members of the set.**

- **Examples:**

- **Vowels in the English alphabet**

$V = \{ a, e, i, o, u \}$

- **First seven prime numbers.**

$X = \{ 2, 3, 5, 7, 11, 13, 17 \}$

Representing Set

Representing a set by:

- 1) Listing (enumerating) the members of the set.
- 2) Definition by property, using the set builder notation $\{x \mid x \text{ has property } P\}$.

Example:

- Even integers between 50 and 63.
 - 1) $E = \{50, 52, 54, 56, 58, 60, 62\}$
 - 2) $E = \{x \mid 50 \leq x < 63, x \text{ is an even integer}\}$

If enumeration of the members is hard we often use ellipses.

Example: a set of integers between 1 and 100

- $A = \{1, 2, 3, \dots, 100\}$

Important set in discrete math

- **Natural numbers:**

- $\mathbb{N} = \{0, 1, 2, 3, \dots\}$

- **Integers**

- $\mathbb{Z} = \{\dots, -2, -1, 0, 1, 2, \dots\}$

- **Positive integers**

- $\mathbb{Z}^+ = \{1, 2, 3, \dots\}$

- **Rational numbers**

- $\mathbb{Q} = \{p / q \mid p \in \mathbb{Z}, q \in \mathbb{Z}, q \neq 0\}$

- **Real numbers**

- $\mathbb{R}$

Equality of Set

Definition: Two sets are equal if and only if they have the same elements.

Example:

- $\{1,2,3\} = \{3,1,2\} = \{1,2,1,3,2\}$

Note: **Duplicates** don't contribute anything new to a set, so remove them. The **order** of the elements in a set doesn't contribute anything new.

Example: Are $\{1,2,3,4\}$ and $\{1,2,2,4\}$ equal?

No!

Universal set

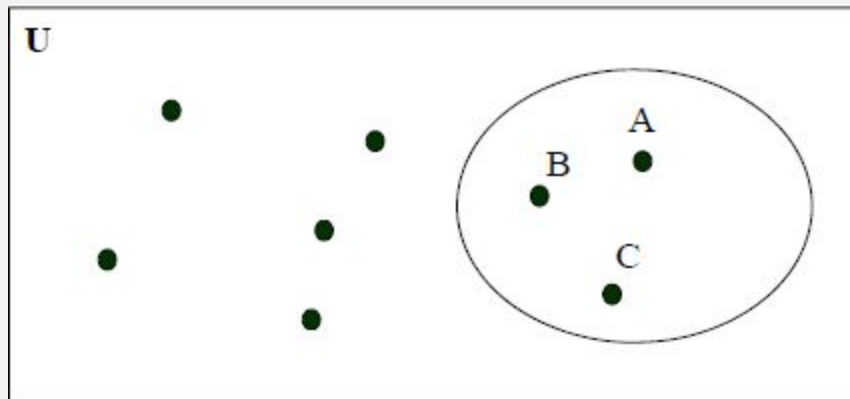
Special sets:

- The **universal set** is denoted by **U**: the set of all objects under the consideration.
- The **empty set** is denoted as $\emptyset$ or $\{ \}$.

Venn Diagram

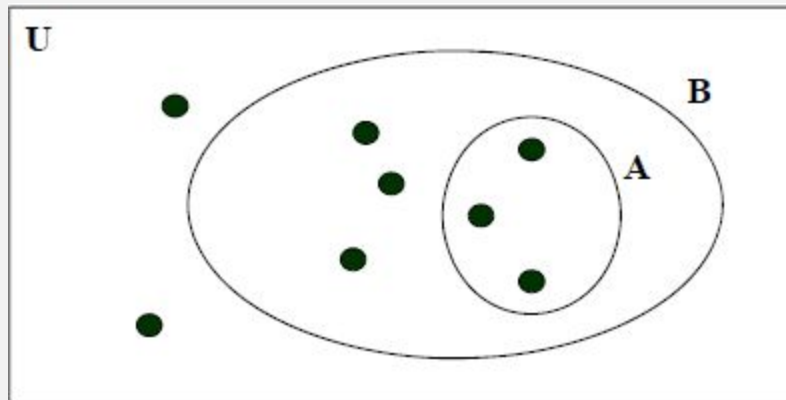
A set can be visualized using **Venn Diagrams**:

– $V = \{ A, B, C \}$



A subset

Definition: A set **A** is said to be a subset of **B** if and only if **every element of A is also an element of B**. We use $A \subseteq B$ to indicate **A is a subset of B**.



Alternate way to define A is a subset of B:

$$\forall x(x \in A) \rightarrow (x \in B)$$

Empty set/subset property

Theorem : $\phi \in S$

- Empty set is a subset of any set.

Proof:

- Recall the definition of a subset: all elements of a set A must be also elements of B:

$$\forall x(x \in A \rightarrow x \in B)$$

- We must show the following implication holds for any

$$\forall x(x \in \phi \rightarrow x \in S)$$

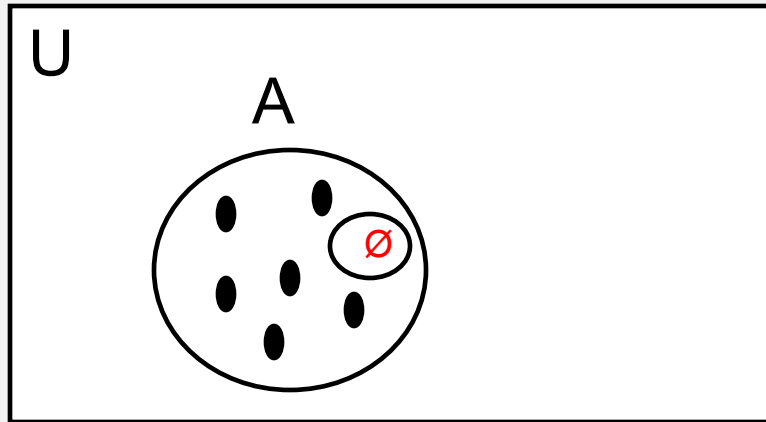
- Since the empty set does not contain any element, $x \in \phi$ is **always False**
- Then the implication is **always True**. $(F \rightarrow T/F = T)$

End of proof

Venn diagram of Empty set

Theorem : $\phi \in S$

- Empty set is a subset of any set.



Subset property

Theorem: $S \subseteq S$

- Any set S is a subset of itself

Proof:

- the definition of a subset says: all elements of a set A must be also elements of B : $\forall x(x \in A \rightarrow x \in B)$

- Applying this to S we get:
- $\forall x(x \in S \rightarrow x \in S)$ which is trivially **True**
- End of proof

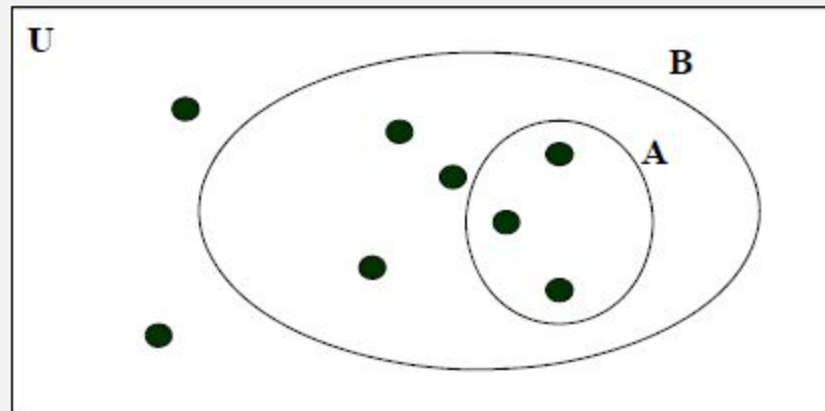
Note on equivalence:

- Two sets are equal if each is a subset of the other set.

A proper Subset

Definition:

A set **A** is said to be a **proper subset** of **B** if and only if $A \subseteq B$ **and** $A \neq B$. We denote that **A** is a proper subset of **B** with the notation $A \subset B$.



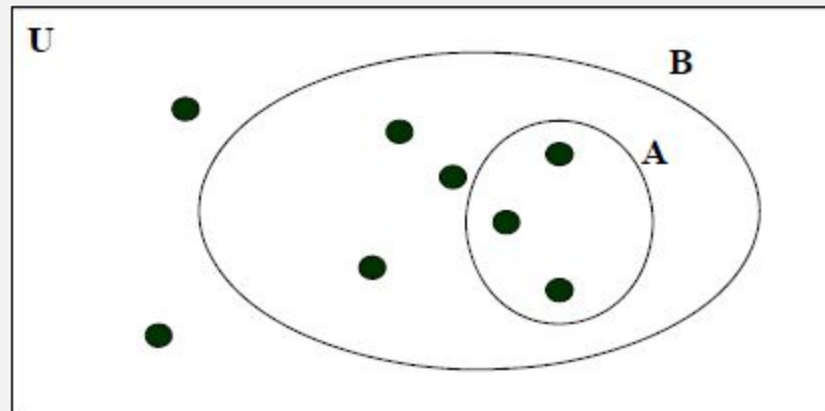
Example: $A = \{1, 2, 3\}$ $B = \{1, 2, 3, 4, 5\}$

Is: $A \subset B$?

A proper Subset

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A set **A** is said to be a **proper subset** of **B** if and only if $A \subseteq B$ **and** $A \neq B$. We denote that **A** is a proper subset of **B** with the notation $A \subset B$.



Example: $A = \{1, 2, 3\}$ $B = \{1, 2, 3, 4, 5\}$

Is: $A \subset B$? **Yes.**

Cardinality

Definition: Let S be a set. If there are **exactly n distinct elements** in S , where **n is a nonnegative integer**, we say S is a finite set and that **n is the cardinality** of S . The cardinality of S is denoted by **$|S|$** .

Examples:

- $V = \{1, 2, 3, 4, 5\}$
 $|V| = 5$
- $A = \{1, 2, 3, 4, \dots, 20\}$
 $|A| = 20$
- $|\emptyset| = 0$

Infinite set

Definition: A set is infinite if it is not finite.

Examples:

- The set of natural numbers is an infinite set.
- $\mathbb{N} = \{1, 2, 3, \dots\}$
- The set of real numbers is an infinite set.

Power set

Definition: Given a set S , the power set of S is the **set of all subsets** of S . The **power set is denoted by $P(S)$** .

Example

- What is the power set of $\emptyset$? $P(\emptyset) = \{\emptyset\}$
- What is the cardinality of $P(\emptyset)$? $|P(\emptyset)| = 1$.

Assume $\{1,2,3\}$

- $P(\{1,2,3\}) = \{\emptyset, \{1\}, \{2\}, \{3\}, \{1,2\}, \{1,3\}, \{2,3\}, \{1,2,3\}\}$
- $|P(\{1,2,3\})| = 8$

If S is a set with $|S| = n$ then $|P(S)| = ?$ **2^n**

N-tuple

Sets are used to represent unordered collections.

- **Ordered-n tuples** are used to **represent an ordered collection**.

Definition: An ordered n -tuple $(x_1, x_2, \dots, x_N)$ is the ordered collection that has x_1 as its first element, x_2 as its second element, ..., and x_N as its N -th element, $N \geq 2$.



Example: Coordinates of a point in the 2-D plane $(12, 16)$

Cartesian Product

Definition: Let S and T be sets. The Cartesian product of S and T , denoted by $S \times T$, is the **set of all ordered pairs** (s,t) , where $s \in S$ and $t \in T$. Hence,

- $S \times T = \{ (s,t) \mid s \in S \wedge t \in T \}.$

Examples:

- $S = \{1,2\}$ and $T = \{a,b,c\}$
- $S \times T = \{ (1,a), (1,b), (1,c), (2,a), (2,b), (2,c) \}$
- $T \times S = \{ (a,1), (a,2), (b,1), (b,2), (c,1), (c,2) \}$
- Note: $S \times T \neq T \times S$!!!!

Cardinality of a Cartesian Product

- $|S \times T| = |S| * |T|.$

Example:

- $A = \{\text{John, Peter, Mike}\}$
- $B = \{\text{Jane, Ann, Laura}\}$

- $A \times B = \{(\text{John, Jane}), (\text{John, Ann}), (\text{John, Laura}), (\text{Peter, Jane}), (\text{Peter, Ann}), (\text{Peter, Laura}), (\text{Mike, Jane}), (\text{Mike, Ann}), (\text{Mike, Laura})\}$

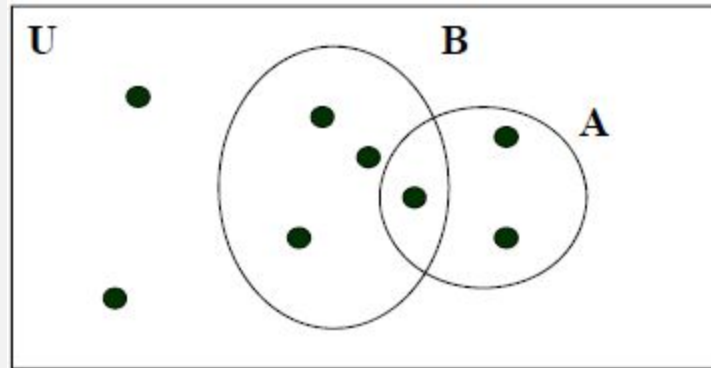
- $|A \times B| = 9$
- $|A|=3, |B|=3 \rightarrow |A| |B|= 9$

Definition: A subset of the Cartesian product $A \times B$ is called a relation from the set A to the set B .

Set Operation

Definition: Let A and B be sets. The **union of A and B**, denoted by **$A \cup B$** , is the set that contains those elements that are in both A and B.

- Alternate: $A \cup B = \{ x \mid x \in A \vee x \in B \}$.



Example:

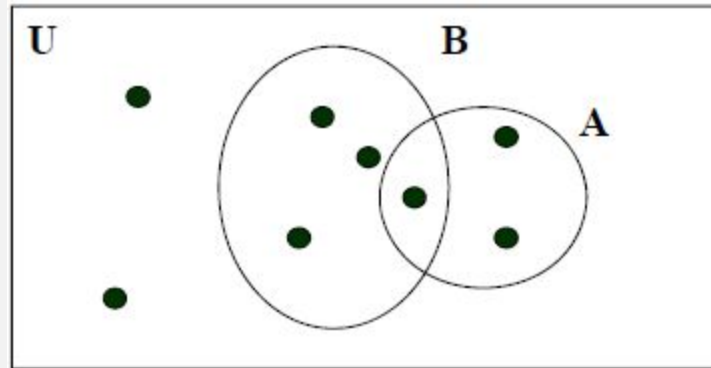
- $A = \{1, 2, 3, 6\}$ and $B = \{2, 4, 6, 9\}$

- **$A \cup B = \{1, 2, 3, 4, 6, 9\}$**

Set Operation

Definition: Let A and B be sets. The **intersection of A and B**, denoted by **$A \cap B$** , is the set that contains those elements that are in both A and B.

- Alternate: **$A \cap B = \{ x \mid x \in A \wedge x \in B \}$** .



Example:

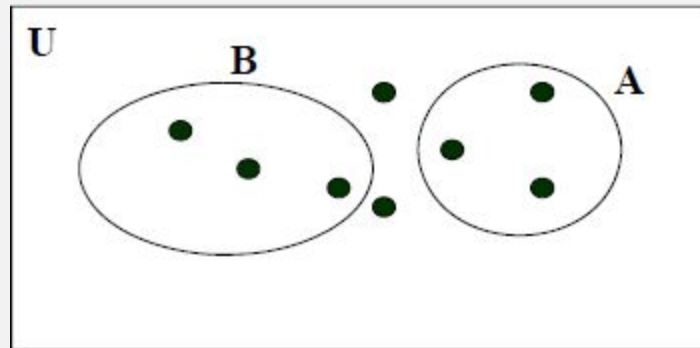
- $A = \{1, 2, 3\}$ and $B = \{2, 4, 6, 9\}$

- **$A \cap B = \{2\}$**

Disjoin Set

Definition: Two sets are called disjoint if their intersection is empty.

- Alternate: A and B are disjoint **if and only if** $A \cap B = \emptyset$.



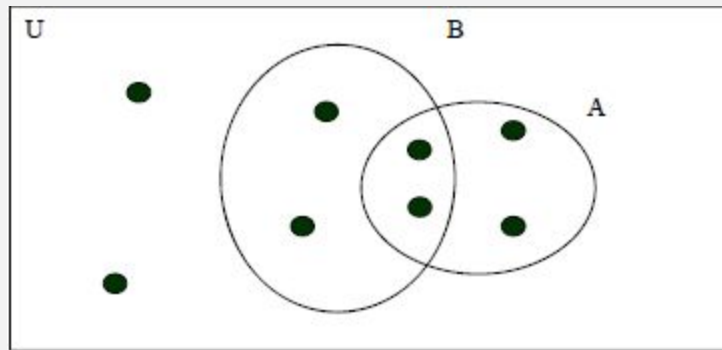
Example:

- $A = \{1, 2, 3, 6\}$ $B = \{4, 7, 8\}$ Are these disjoint?
- Yes.
- $A \cap B = \emptyset$

Cardinality of set union

Cardinality of the set union.

- $|A \cup B| = |A| + |B| - |A \cap B|$

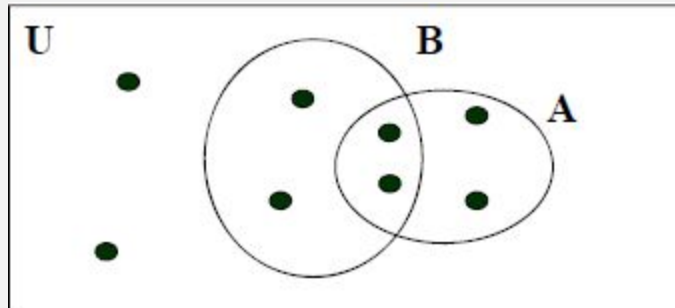


Why this formula? Correct for an over-count.

Set Difference

Definition: Let A and B be sets. The difference of A and B, denoted by $A - B$, is the set containing those **elements that are in but not in B**. The difference of A and B is also called the complement of B with respect to A.

- Alternate: $A - B = \{x \mid x \in A \wedge x \notin B\}$



Example: $A = \{1, 2, 3, 5, 7\}$ $B = \{1, 5, 6, 8\}$

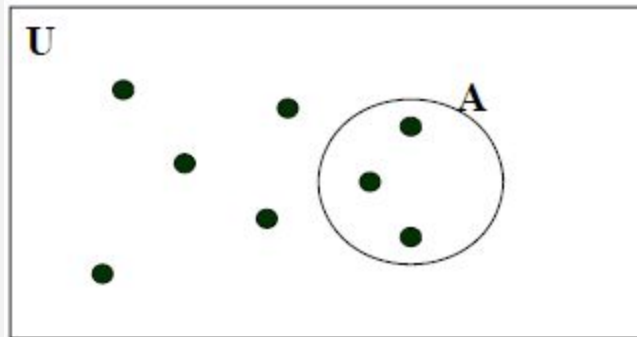
- $A - B = \{2, 3, 7\}$

Complement of a Set

Definition: Let U be the universal set: the set of all objects under the consideration.

Definition: The complement of the set A , denoted by $\tilde{A}$, is the complement of A with respect to U .

• Alternate: Alternate: $\overline{A} = \{x \mid x \notin A\}$



Example: $U = \{1, 2, 3, 4, 5, 6, 7, 8\}$ $A = \{1, 3, 5\}$

• $\tilde{A} = \{2, 4, 6, 7, 8\}$

Generalized union

Definition: The union of a collection of sets is the set that contains those elements that are **members of at least one set** in the collection.

$$\bigcup_{i=1}^n A_i = \{A_1 \cup A_2 \cup \dots \cup A_n\}$$

Example:

- Let $A_i = \{1, 2, \dots, i\}$ $i = 1, 2, \dots, n$

$$\bigcup_{i=1}^n A_i = \{1, 2, \dots, n\}$$

$$\begin{aligned} A_1 &= \{1\} \\ A_2 &= \{1, 2\} \\ A_3 &= \{1, 2, 3\} \end{aligned}$$

.....

.....

$$A_n = \{1, 2, 3, 4, \dots, n\}$$

Generalized intersection

Definition: The intersection of a collection of sets is the set that contains those elements **that are members of all sets in the collection.**

$$\bigcap_{i=1}^n A_i = \{A_1 \cap A_2 \cap \dots \cap A_n\}$$

Example:

- Let $A_i = \{1, 2, \dots, i\}$ $i = 1, 2, \dots, n$

$$\bigcap_{i=1}^n A_i = \{1\}$$

$$A_1 = \{1\}$$

$$A_2 = \{1, 2\}$$

$$A_3 = \{1, 2, 3\}$$

.....

.....

$$A_n = \{1, 2, 3, 4, \dots, n\}$$

Binary relation

Definition: Let A and B be two sets. A **binary relation** from A to B is **a subset** of a Cartesian product **$A \times B$** .

- Let $R \subseteq A \times B$ means R is a set of ordered pairs of the form **(a,b) where $a \in A$ and $b \in B$** .
- We use the notation **$a R b$** to denote $(a,b) \in R$ and **$a R b$** to denote $(a,b) \in R$. If **$a R b$** , we say a is related to b by R .

Example: Let $A=\{a,b,c\}$ and $B=\{1,2,3\}$.

- **Is $R=\{(a,1),(b,2),(c,2)\}$ a relation from A to B ?**

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- Is $R=\{(a,1),(b,2),(c,2)\}$ a relation from A to B ? **Yes.**
- **Is $Q=\{(1,a),(2,b)\}$ a relation from A to B ?**

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Definition: Let A and B be two sets. A **binary relation** from A to B is **a subset** of a Cartesian product **$A \times B$** .

- Let $R \subseteq A \times B$ means R is a set of ordered pairs of the form (a,b) where $a \in A$ and $b \in B$.
- We use the notation **$a R b$** to denote $(a,b) \in R$ and **$a R b$** to denote $(a,b) \in R$. If $a R b$, we say a is related to b by R .

Example: Let $A=\{a,b,c\}$ and $B=\{1,2,3\}$.

- Is $R=\{(a,1),(b,2),(c,2)\}$ a relation from A to B ? **Yes.**
- Is $Q=\{(1,a),(2,b)\}$ a relation from A to B ? **No.**
- Is $P=\{(a,a),(b,c),(b,a)\}$ a relation from A to A ?

Binary relation

Definition: Let A and B be two sets. A **binary relation** from A to B is **a subset** of a Cartesian product **$A \times B$** .

- Let $R \subseteq A \times B$ means R is a set of ordered pairs of the form (a,b) where $a \in A$ and $b \in B$.
- We use the notation **$a R b$** to denote $(a,b) \in R$ and **$a R b$** to denote $(a,b) \in R$. If $a R b$, we say a is related to b by R .

Example: Let $A=\{a,b,c\}$ and $B=\{1,2,3\}$.

- Is $R=\{(a,1),(b,2),(c,2)\}$ a relation from A to B ? **Yes.**
- Is $Q=\{(1,a),(2,b)\}$ a relation from A to B ? **No.**
- Is $P=\{(a,a),(b,c),(b,a)\}$ a relation from A to A ? **Yes**

Representing Binary relation

We can graphically represent a binary relation R as follows:

- if $a R b$ then draw an arrow from a to b .

$a \rightarrow b$

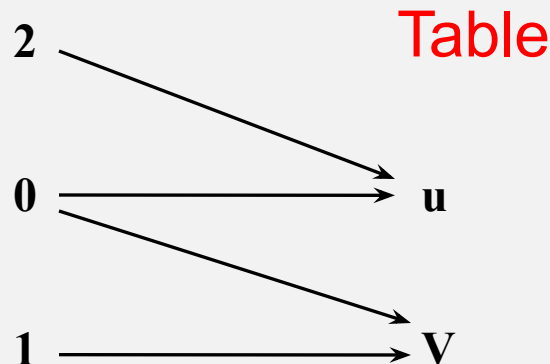
Example:

- Let $A = \{0, 1, 2\}$, $B = \{u, v\}$ and

$$R = \{ (0, u), (0, v), (1, v), (2, u) \}$$

- Note: $R \subseteq A \times B$

- **Graph:**



	u	v
0	x	x
1		x
2	x	

Reflexive Relation

Definition (reflexive relation) : A relation R on a set A is called reflexive if $(a,a) \in R$ for every element $a \in A$.

Example 1:

- Assume relation $R_{\text{div}} = \{(a, b), \text{ if } a \mid b\}$ on $A = \{1, 2, 3, 4\}$

$$R_{\text{div}} = \{(1,1), (1,2), (1,3), (1,4), (2,2), (2,4), (3,3), (4,4)\}$$

- **Is R_{div} reflexive?**

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$$R_{\text{div}} = \{(1,1), (1,2), (1,3), (1,4), (2,2), (2,4), (3,3), (4,4)\}$$

- **Is R_{div} reflexive?**

- **Answer: Yes.** $(1,1), (2,2), (3,3), \text{ and } (4,4) \in R_{\text{div}}$

Reflexive Relation

Definition (reflexive relation) : A relation R on a set A is called reflexive if $(a,a) \in R$ for every element $a \in A$.

Example 2:

- Relation R_{fun} on $A = \{1,2,3,4\}$ defined as:
- $R_{\text{fun}} = \{(1,2), (2,2), (3,3)\}$.
- **Is R_{fun} reflexive?**

Reflexive Relation

Definition (reflexive relation) : A relation R on a set A is called reflexive if $(a,a) \in R$ for every element $a \in A$.

Example 2:

- Relation R_{fun} on $A = \{1,2,3,4\}$ defined as:
- $R_{\text{fun}} = \{(1,2), (2,2), (3,3)\}$.
- **Is R_{fun} reflexive?**
- **No.** It is not reflexive since $(1,1), (4,4) \notin R_{\text{fun}}$.

Symmetric Relation

Definition (symmetric relation): A relation R on a set A is called **symmetric** if

$$\forall a, b \in A (a, b) \in R \rightarrow (b, a) \in R$$

Example 1:

- $R_{\text{div}} = \{(a, b), \text{ if } a \mid b\}$ on $A = \{1, 2, 3, 4\}$
- $R_{\text{div}} = \{(1, 1), (1, 2), (1, 3), (1, 4), (2, 2), (2, 4), (3, 3), (4, 4)\}$
- **Is R_{div} symmetric?**

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- $R_{\text{div}} = \{(1, 1), (1, 2), (1, 3), (1, 4), (2, 2), (2, 4), (3, 3), (4, 4)\}$
- **Is R_{div} symmetric?**
- **Answer: No.** It is not symmetric since $(1, 2) \in R$ but $(2, 1) \notin R$.

Symmetric Relation

Definition (symmetric relation): A relation R on a set A is called **symmetric** if

$$\forall a, b \in A (a, b) \in R \rightarrow (b, a) \in R$$

Example 2:

- $R_{\neq}$ on $A=\{1,2,3,4\}$, such that $a R_{\neq} b$ if and only if $a \neq b$.

$$R_{\neq} = \{(1,2), (1,3), (1,4), (2,1), (2,3), (2,4), (3,1), (3,2), (3,4), (4,1), (4,2), (4,3)\}$$

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- **Is $R_{\neq}$ symmetric ?**

Answer: Yes. $\forall a, b \in A (a, b) \in R \rightarrow (b, a) \in R$

Transitive Relation

Definition (transitive relation): A relation R on a set A is called transitive if

$$[(a,b) \in R \text{ and } (b,c) \in R] \rightarrow (a,c) \in R \quad \forall a,b,c \in A$$

- **Example 1:**

- $R_{\text{div}} = \{(a,b) \mid a \mid b\}$ on $A = \{1,2,3,4\}$

- $R_{\text{div}} = \{(1,1), (1,2), (1,3), (1,4), (2,2), (2,4), (3,3), (4,4)\}$

- **Is R_{div} transitive?**

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- **Is R_{div} transitive?**
- **Answer: Yes.**

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- Example 2:
- $R_{\neq}$ on $A=\{1,2,3,4\}$, such that $a R_{\neq} b$ if and only if $a \neq b$.
- $R_{\neq} = \{(1,2), (1,3), (1,4), (2,1), (2,3), (2,4), (3,1), (3,2), (3,4), (4,1), (4,2), (4,3)\}$
- Is $R_{\neq}$ transitive ?

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- Is $R_{\neq}$ transitive ?

- Answer: **No**. It is not transitive since $(1,2) \in R$ and $(2,1) \in R$ but $(1,1)$ is not an element of R .

Combining Relation

Definition: Let A and B be sets. A binary relation from A to B is a subset of a Cartesian product $A \times B$.

- Let $R \subseteq A \times B$ means R is a set of ordered pairs of the form (a,b) where $a \in A$ and $b \in B$.

Combining Relations

- Relations are **sets** $\rightarrow$ combinations via **set operations**
- Set operations of: **union, intersection, difference and symmetric difference.**

Combining Relation

Example:

- Let $A = \{1,2,3\}$ and $B = \{u,v\}$ and
- $R1 = \{(1,u), (2,u), (2,v), (3,u)\}$
- $R2 = \{(1,v), (3,u), (3,v)\}$

What is:

- $R1 \cup R2 = ?$

Combining Relation

Example:

- Let $A = \{1,2,3\}$ and $B = \{u,v\}$ and
- $R1 = \{(1,u), (2,u), (2,v), (3,u)\}$
- $R2 = \{(1,v), (3,u), (3,v)\}$

What is:

- $R1 \cup R2 = \{(1,u), (1,v), (2,u), (2,v), (3,u), (3,v)\}$
- $R1 \cap R2 = ?$

Combining Relation

Example:

- Let $A = \{1,2,3\}$ and $B = \{u,v\}$ and
- $R1 = \{(1,u), (2,u), (2,v), (3,u)\}$
- $R2 = \{(1,v), (3,u), (3,v)\}$

What is:

- $R1 \cup R2 = \{(1,u), (1,v), (2,u), (2,v), (3,u), (3,v)\}$
- $R1 \cap R2 = \{(3,u)\}$
- $R1 - R2 = ?$

Combining Relation

Example:

- Let $A = \{1,2,3\}$ and $B = \{u,v\}$ and
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What is:

- $R1 \cup R2 = \{(1,u), (1,v), (2,u), (2,v), (3,u), (3,v)\}$
- $R1 \cap R2 = \{(3,u)\}$
- $R1 - R2 = \{(1,u), (2,u), (2,v)\}$
- $R2 - R1 = ?$

Combining Relation

Example:

- Let $A = \{1,2,3\}$ and $B = \{u,v\}$ and
- $R1 = \{(1,u), (2,u), (2,v), (3,u)\}$
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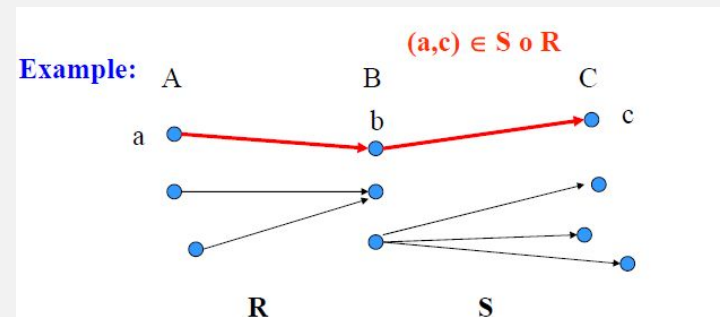
Composite Relation

Definition: Let R be a relation from a set A to a set B and S a relation from B to a set C . The **composite of R and S** is the relation consisting of the ordered pairs (a,c) where $a \in A$ and $c \in C$, and for which there is a $b \in B$ such that $(a,b) \in R$ and $(b,c) \in S$. We denote the composite of R and S by $S \circ R$.

Examples:

- Let $A = \{1,2,3\}$, $B = \{0,1,2\}$ and $C = \{a,b\}$.
- $R = \{(1,0), (1,2), (3,1), (3,2)\}$
- $S = \{(0,b), (1,a), (2,b)\}$

- **$S \circ R = ?$**

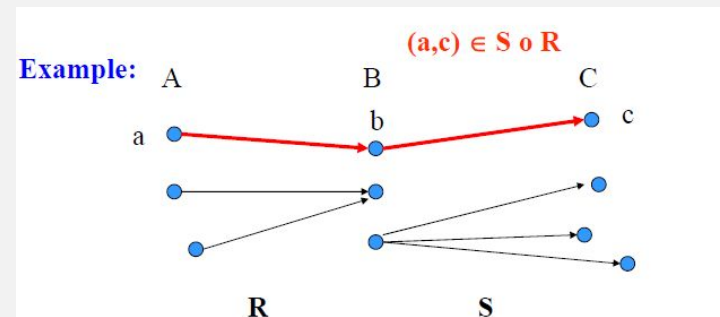


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Examples:

- Let $A = \{1,2,3\}$, $B = \{0,1,2\}$ and $C = \{a,b\}$.
- $R = \{(1,0), (1,2), (3,1), (3,2)\}$
- $S = \{(0,b), (1,a), (2,b)\}$
- **$S \circ R = \{(1,b), (3,a), (3,b)\}$**



Composite Relation

Definition: Let R be a relation on a set A . The **powers** R^n , $n = 1, 2, 3, \dots$ is defined inductively by

- $R^1 = R$ **and** $R^{n+1} = R^n \circ R$.

Examples

- $R = \{(1,2), (2,3), (2,4), (3,3)\}$ is a relation on $A = \{1,2,3,4\}$.
- **$R^1 = ?$**

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- $R^1 = \{(1,2), (2,3), (2,4), (3,3)\}$
- $R^2 = ?$

Composite Relation

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- $R^3 = ?$

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- $R^3 = \{(1,3), (2,3), (3,3)\}$
- $R^4 =$

Composite Relation

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- $R^2 = \{(1,3), (1,4), (2,3), (3,3)\}$
- $R^3 = \{(1,3), (2,3), (3,3)\}$
- $R^4 = \{(1,3), (2,3), (3,3)\}$
- $R^k = ?$ when $k > 3$

Composite Relation

Definition: Let R be a relation on a set A . The **powers** R^n , $n = 1, 2, 3, \dots$ is defined inductively by

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Examples

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- $R^k = \{(1,3), (2,3), (3,3)\}$

Transitive Relation

Definition (transitive relation): A relation R on a set A is called **transitive** if

• $[(a,b) \in R \text{ and } (b,c) \in R] \rightarrow (a,c) \in R$ for all $a, b, c \in A$.

• **Example 1:**

• $R_{\text{div}} = \{(a, b), \text{ if } a \mid b\}$ on $A = \{1, 2, 3, 4\}$

• $R_{\text{div}} = \{(1,1), (1,2), (1,3), (1,4), (2,2), (2,4), (3,3), (4,4)\}$

• **Is R_{div} transitive?**

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• **Is R_{div} transitive?**

• **Answer: Yes.**

Equivalence Relation

Definition: A relation R on a set A is called an **equivalence relation** if it is **reflexive**, **symmetric** and **transitive**.

Example: Let $A = \{0,1,2,3,4,5,6\}$ and

- $R = \{(a,b) \mid a,b \in A, a \equiv b \pmod{3}\}$ (a is congruent to b modulo 3)

Congruencies:

- $0 \pmod{3} = 0$ $1 \pmod{3} = 1$ $2 \pmod{3} = 2$ $3 \pmod{3} = 0$
- $4 \pmod{3} = 1$ $5 \pmod{3} = 2$ $6 \pmod{3} = 0$

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Relation R has the following pairs:

- $(0,0)$ $(0,3), (3,0), (0,6), (6,0)$
- $(3,3), (3,6), (6,3), (6,6)$ $(1,1), (1,4), (4,1), (4,4)$
- $(2,2), (2,5), (5,2), (5,5)$

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Is R reflexive

Equivalence Relation

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Is R reflexive Yes

Is R symmetric

Equivalence Relation

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Is R reflexive Yes

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Is R transitive

Equivalence Relation

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Is R reflexive Yes

Is R symmetric Yes

Is R transitive Yes

Then

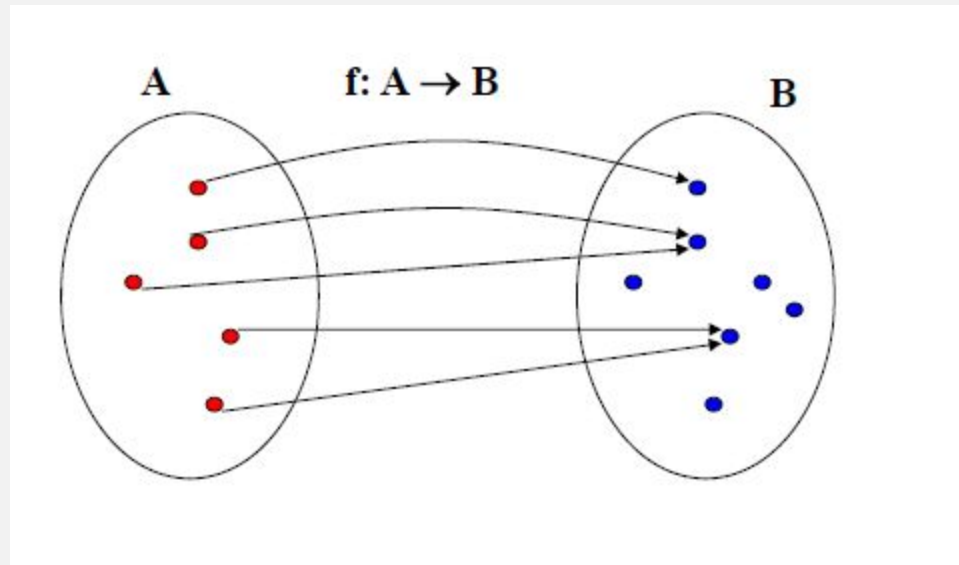
• R is an equivalence relation

Equivalence Relation - proof

- $i = j \bmod 3$ ($i-j$ is a multiple of 3) is an equivalence relation on natural numbers
 - Reflexive?
 - Yes, $i-i = 0$ which is a multiple of 3
 - Symmetric?
 - Yes, if $i-j$ is a multiple of 3 so is $j-i$
 - Transitive?
 - Yes, if $i-j = m*3$ and $j-k = n*3$ then $i-k = i-j - (j-k) = (m-n)*3$

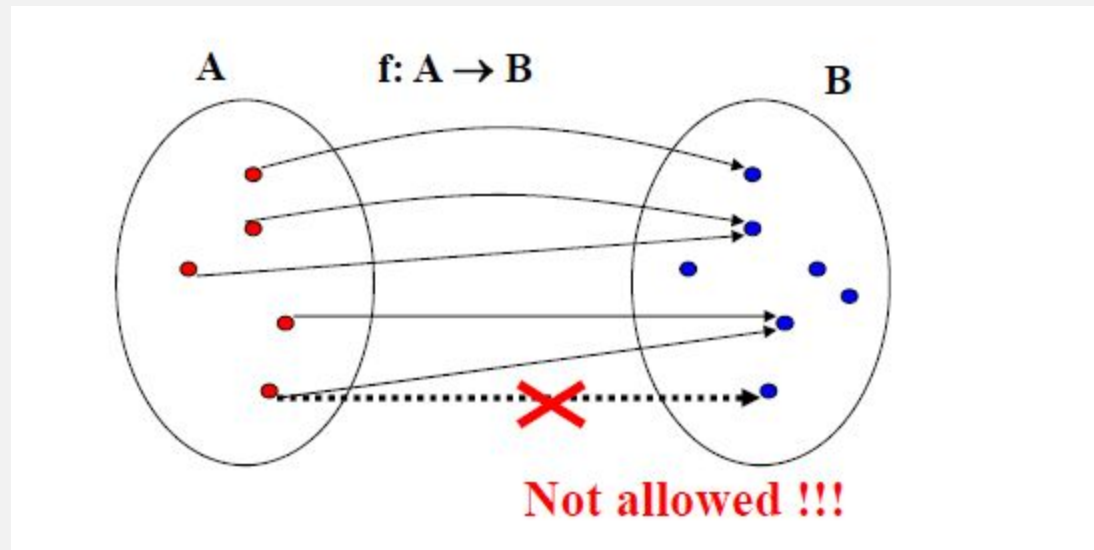
Function

Definition: Let A and B be two sets. A function from A to B , denoted $f : A \rightarrow B$, is an assignment of exactly one element of B to each element of A . We write $f(a) = b$ to denote the assignment of b to an element a of A by the function f .



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Representing Function

Representations of functions:

1. Explicitly state the assignments in between elements of the two sets
2. Compactly by a formula. (using 'standard' functions)

Example1:

- Let $A = \{1,2,3\}$ and $B = \{a,b,c\}$
- Assume f is defined as:
 - $1 \rightarrow c$
 - $2 \rightarrow a$
 - $3 \rightarrow c$
- Is f a function ?
- Yes. since $f(1)=c$, $f(2)=a$, $f(3)=c$. each element of A is assigned an element from B

Representing Function

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- Is g a function ?

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- Assume g is defined as:
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 - $2 \rightarrow a$
 - $3 \rightarrow c$
- Is g a function ?
- No. since $g(1)$ is assigned both c and b .

Representing Function

Representations of functions:

1. Explicitly state the assignments in between elements of the two sets
2. Compactly by a formula. (using 'standard' functions)

Example 3:

- $A = \{0, 1, 2, 3, 4, 5, 6, 7, 8, 9\}$, $B = \{0, 1, 2\}$
- Define $h: A \rightarrow B$ as:
$$h(x) = x \bmod 3.$$
- (the result is the remainder after the division by 3)
- Assignments:
 - $0 \rightarrow 0$ $3 \rightarrow 0$
 - $1 \rightarrow 1$ $4 \rightarrow 1$
 - $2 \rightarrow 2$...

Notation of Set

Definitions: Let f be a function from A to B .

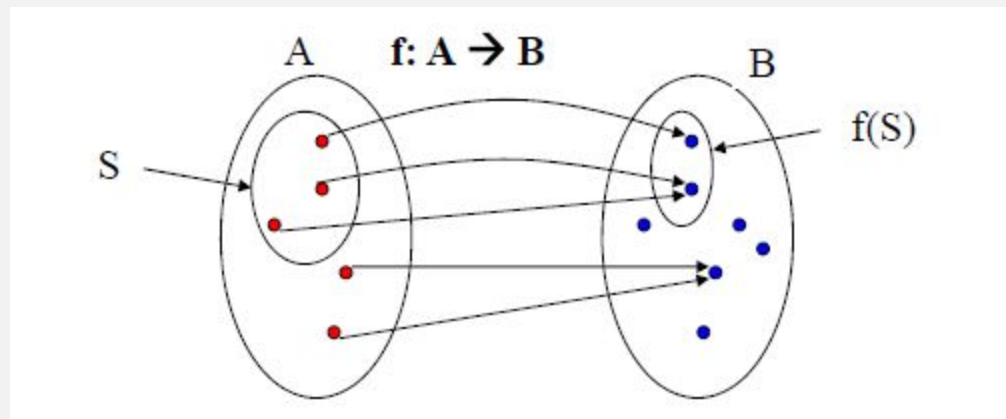
- We say that A is the **domain** of f and B is the **codomain** of f .
- If $f(a) = b$, b is the **image** of a and a is a **pre-image** of b .
- The **range** of f is the **set of all images of elements of A** . Also, if f is a function from A to B , we say f maps A to B .

Example: Let $A = \{1,2,3\}$ and $B = \{a,b,c\}$

- Assume f is defined as: $1 \rightarrow c, 2 \rightarrow a, 3 \rightarrow c$
- What is the image of 1?
- $1 \rightarrow c$ **c is the image of 1**
- What is the pre-image of a ?
- $2 \rightarrow a$ **2 is a pre-image** of a .
- Domain of f ? $\{1,2,3\}$
- Codomain of f ? $\{a,b,c\}$
- **Range** of f ? $\{a,c\}$

Image of subset

Definition: Let f be a function from set A to set B and let S be a subset of A . The image of S is a subset of B that consists of the images of the elements of S . We denote the image of S by $f(S)$, so that $f(S) = \{ f(s) \mid s \in S \}$.

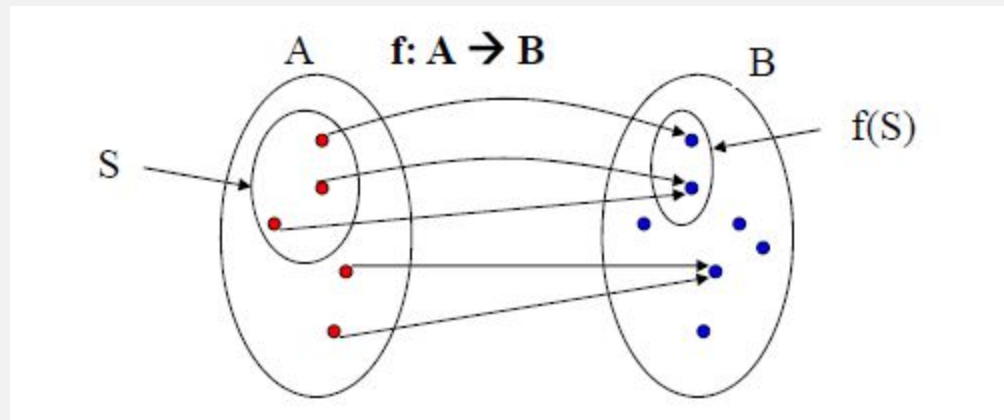


Example:

- Let $A = \{1, 2, 3\}$ and $B = \{a, b, c\}$ and $f: 1 \rightarrow c, 2 \rightarrow a, 3 \rightarrow c$
- Let $S = \{1, 3\}$ then image $f(S) = ?$

Image of subset

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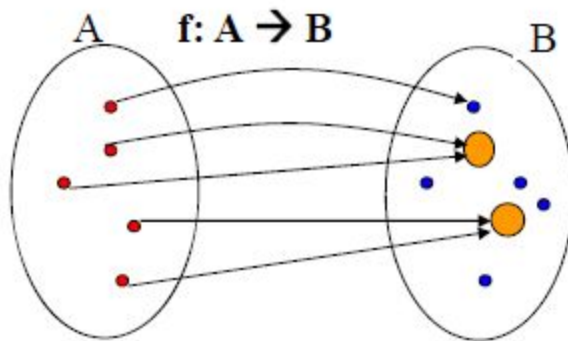
Example:

- Let $A = \{1,2,3\}$ and $B = \{a,b,c\}$ and $f: 1 \rightarrow c, 2 \rightarrow a, 3 \rightarrow c$
- Let $S = \{1,3\}$ then image $f(S) = \{c\}$

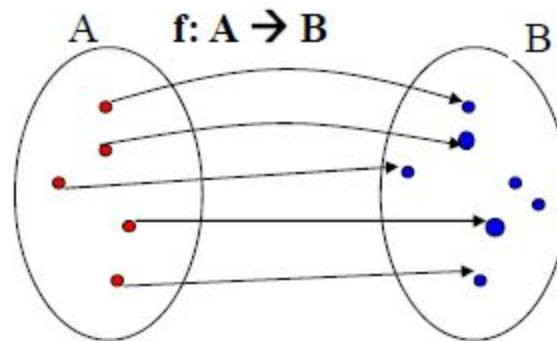
Injective function

Definition: A function f is said to be **one-to-one**, or **injective**, if and only if $f(x) = f(y)$ implies $x = y$ for all x, y in the domain of f . A function is said to be **an injection** if it is **one-to-one**.

A **function** for which **every element of the range of the function** corresponds to **exactly one element of the domain**.



Not injective



Injective function

Injective function

Example 1: Let $A = \{1,2,3\}$ and $B = \{a,b,c\}$

- Define f as

- $1 \rightarrow c$

- $2 \rightarrow a$

- $3 \rightarrow c$

- Is f one to one?

Injective function

Example 1: Let $A = \{1,2,3\}$ and $B = \{a,b,c\}$

- Define f as
 - $1 \rightarrow c$
 - $2 \rightarrow a$
 - $3 \rightarrow c$
- Is f one to one?
- No, it is not one-to-one
- since $f(1) = f(3) = c$, and $1 \neq 3$.

Example 2: Let $g : \mathbb{Z} \rightarrow \mathbb{Z}$, where $g(x) = 2x - 1$.

- Is g is one-to-one?

Injective function

Example 1: Let $A = \{1,2,3\}$ and $B = \{a,b,c\}$

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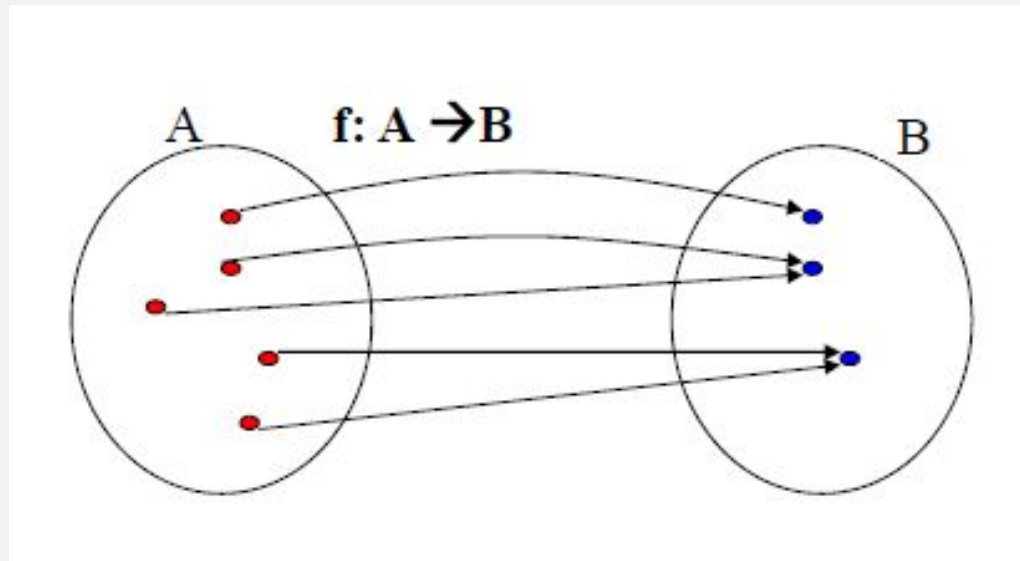
Example 2: Let $g : \mathbb{Z} \rightarrow \mathbb{Z}$, where $g(x) = 2x - 1$.

- Is g is one-to-one?
- Yes.
- Why? $g(a) = g(b)$, i.e., $2a - 1 = 2b - 1 \Rightarrow 2a = 2b \Rightarrow a = b$.

Surjective function

Definition: A function f from A to B is called onto, or surjective, if and only if **for every** $b \in B$ there is an element $a \in A$ such that **$f(a) = b$** .

Alternative: **all co-domain** elements are covered



Surjective function

Example 1: Let $A = \{1,2,3\}$ and $B = \{a,b,c\}$

– Define f as

- $1 \rightarrow c$
- $2 \rightarrow a$
- $3 \rightarrow c$
- Is f an onto?

Surjective function

Example 1: Let $A = \{1,2,3\}$ and $B = \{a,b,c\}$

– Define f as

- $1 \rightarrow c$
- $2 \rightarrow a$
- $3 \rightarrow c$
- Is f an onto?
- **No.** f is not onto, since $b \in B$ has no pre-image.

Example 2: $A = \{0,1,2,3,4,5,6,7,8,9\}$, $B = \{0,1,2\}$

– Define $h: A \rightarrow B$ as $h(x) = x \bmod 3$.

- Is h an onto function?

Surjective function

Example 1: Let $A = \{1,2,3\}$ and $B = \{a,b,c\}$

– Define f as

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- $2 \rightarrow a$
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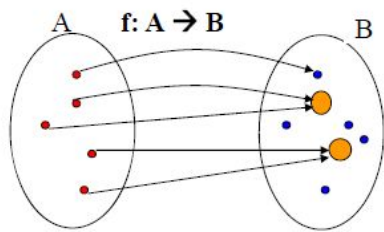
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– Define $h: A \rightarrow B$ as $h(x) = x \bmod 3$.

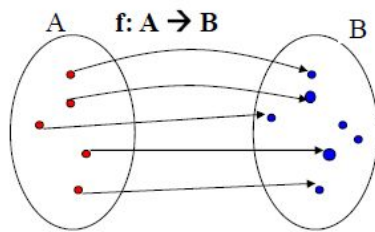
- Is h an onto function?
- **Yes.** h is onto since a pre-image of 0 is 6, a pre-image of 1 is 4, a pre-image of 2 is 8.

Bijjective function

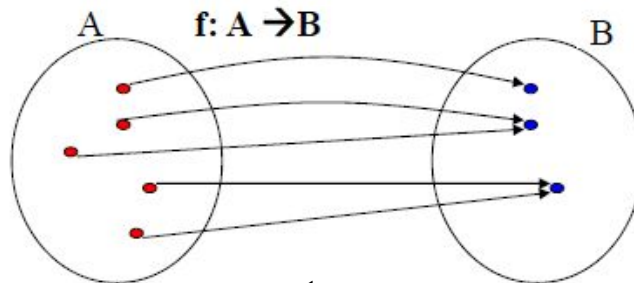
Definition: A function f is called a bijection if it is **both one-to one and onto**.



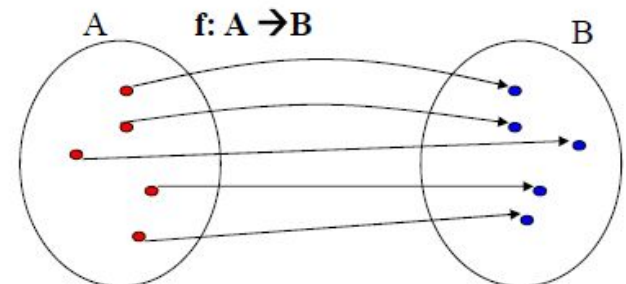
Not injective



Injective function



onto



One to one and onto

Bijjective function

Example 1:

- Let $A = \{1,2,3\}$ and $B = \{a,b,c\}$
 - Define f as
 - $1 \rightarrow c$
 - $2 \rightarrow a$
 - $3 \rightarrow b$
- Is f is a bijection?

Bijjective function

Example 1:

- Let $A = \{1,2,3\}$ and $B = \{a,b,c\}$
 - Define f as
 - $1 \rightarrow c$
 - $2 \rightarrow a$
 - $3 \rightarrow b$
 - Is f is a bijection? Yes. It is both one-to-one and onto.
- Note: Let f be a function from a set A to itself, where A is finite. f is one-to-one if and only if f is onto.
- This is not true if A an infinite set. Define $f : \mathbb{Z} \rightarrow \mathbb{Z}$, where $f(z) = 2 * z$. f is one-to-one but not onto (3 has no pre-image).

Bijjective function

Example 2:

- Define $g : W \rightarrow W$ (whole numbers), where $g(n) = \lfloor n/2 \rfloor$ (floor function).
- $0 \rightarrow \lfloor 0/2 \rfloor = \lfloor 0 \rfloor = 0$
- $1 \rightarrow \lfloor 1/2 \rfloor = \lfloor 0.5 \rfloor = 0$
- $2 \rightarrow \lfloor 2/2 \rfloor = \lfloor 1 \rfloor = 1$
- $3 \rightarrow \lfloor 3/2 \rfloor = \lfloor 1.5 \rfloor = 1$
- ...
- Is g a bijection?
 - No. g is onto but not 1-1 ($g(0) = g(1) = 0$ however $0 \neq 1$).

Bijjective function

Theorem: Let f be a function $f: A \rightarrow A$ from a set A to itself, where A is finite. Then f is one-to-one if and only if f is onto.

Assume

- A is finite and f is one-to-one (injective)
- Is f an onto function (surjection)?.

Bijjective function

Theorem: Let f be a function $f: A \rightarrow A$ from a set A to itself, where A is finite. Then f is one-to-one if and only if f is onto.

Proof:

→ A is finite and f is one-to-one (injective)

- Is f an onto function (surjection)?
- **Yes.** Every element points to exactly one element. **Injection assures they are different.** So we have $|A|$ different elements A points to. Since $f: A \rightarrow A$ the **co-domain is covered** thus the function is also a surjection (and a bijection)

← A is finite and f is an onto function

- **Is the function one-to-one?**

Bijjective function

Theorem: Let f be a function $f: A \rightarrow A$ from a set A to itself, where A is finite. Then f is one-to-one if and only if f is onto.

← A is finite and f is an onto function

- Is the function one-to-one?

Yes. Every element maps to exactly one element and all elements in A are covered. Thus the mapping must be one-to-one

Bijjective function

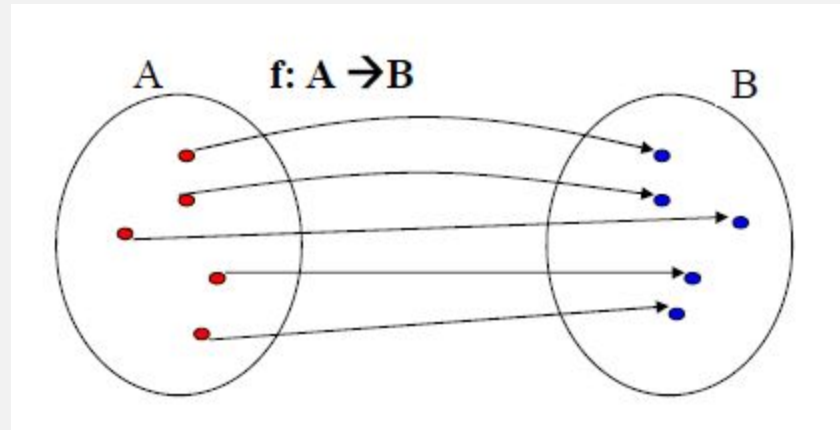
Theorem. Let f be a function from a set A to itself, where A is finite. Then f is one-to-one if and only if f is onto.

Please note the above is **not true when A is an infinite** set.

- Example:
 - $f : \mathbb{Z} \rightarrow \mathbb{Z}$, where $f(z) = 2 * z$.
 - f is one-to-one but not onto.
- $1 \rightarrow 2$
- $2 \rightarrow 4$
- $3 \rightarrow 6$
- **3 has no pre-image.**

Bijjective Function

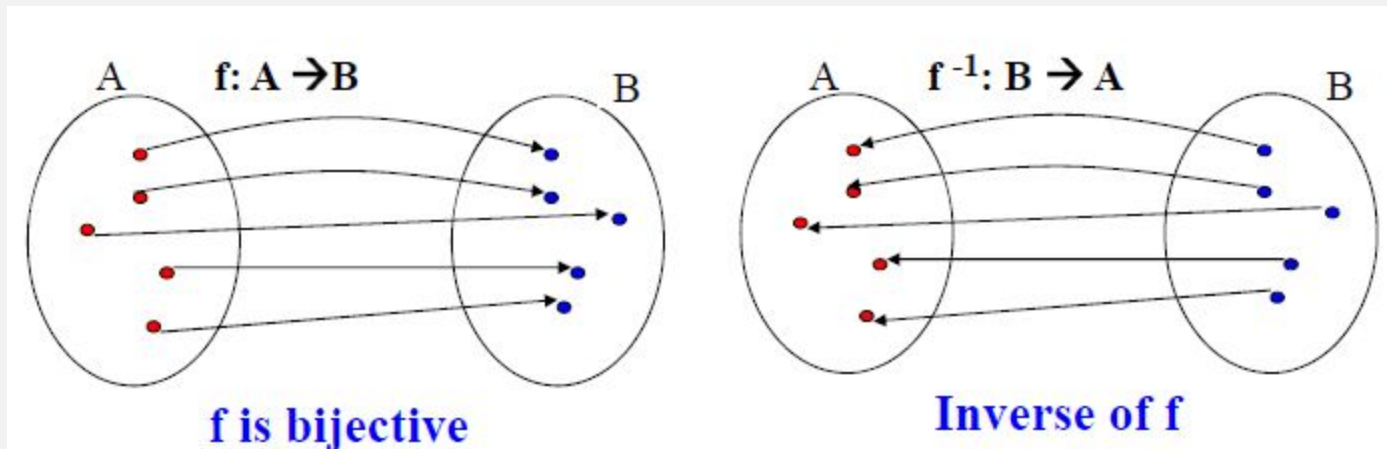
Definition: A function f is called a bijection if it is **both** one-to-one **and** onto.



Inverse Function

Definition: Let f be a **bijection** from set A to set B . **The inverse function of f** is the function that assigns to an element b from B the unique element a in A such that $f(a) = b$.

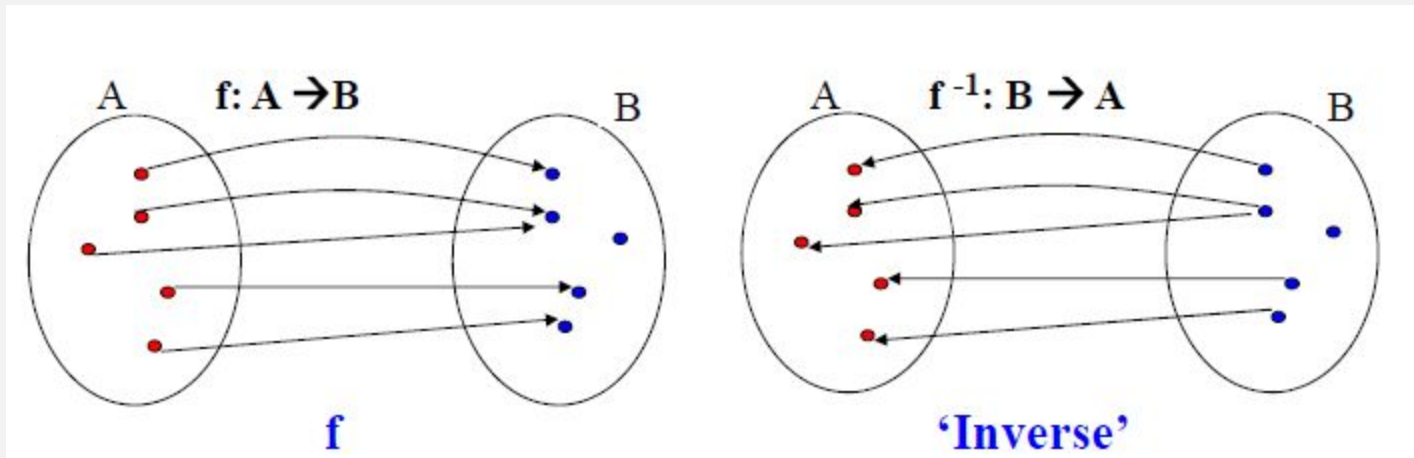
The inverse function of f is denoted by f^{-1} . Hence, $f^{-1}(b) = a$, when $f(a) = b$. If the inverse function of f exists, f is called invertible.



Inverse Function

Note: if f is not a bijection then it is not possible to define the inverse function of f . **Why?**

Assume f is **not one-to-one**:
?

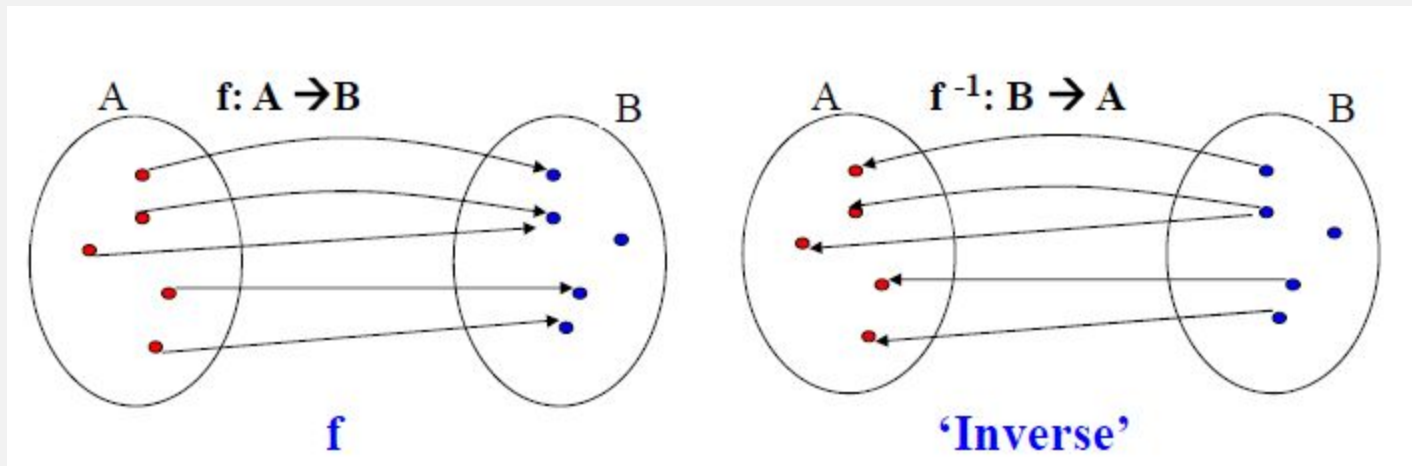


Inverse Function

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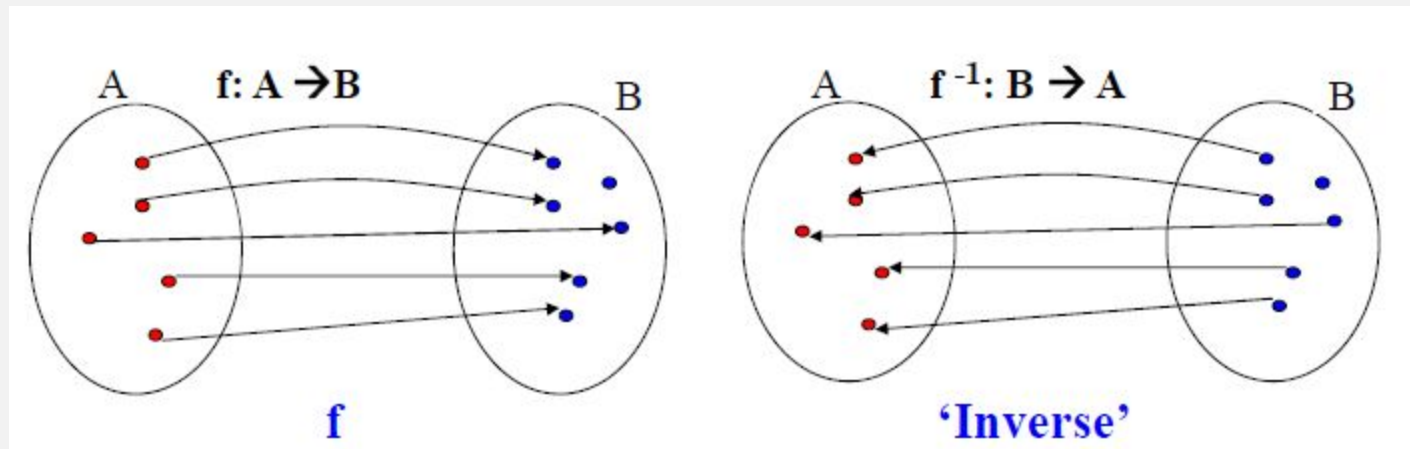
Inverse is **not a function**. One element of B is mapped to two different elements.



Inverse Function

Note: if f is not a bijection then it is not possible to define the inverse function of f . **Why?**

Assume f is **not onto**:
?

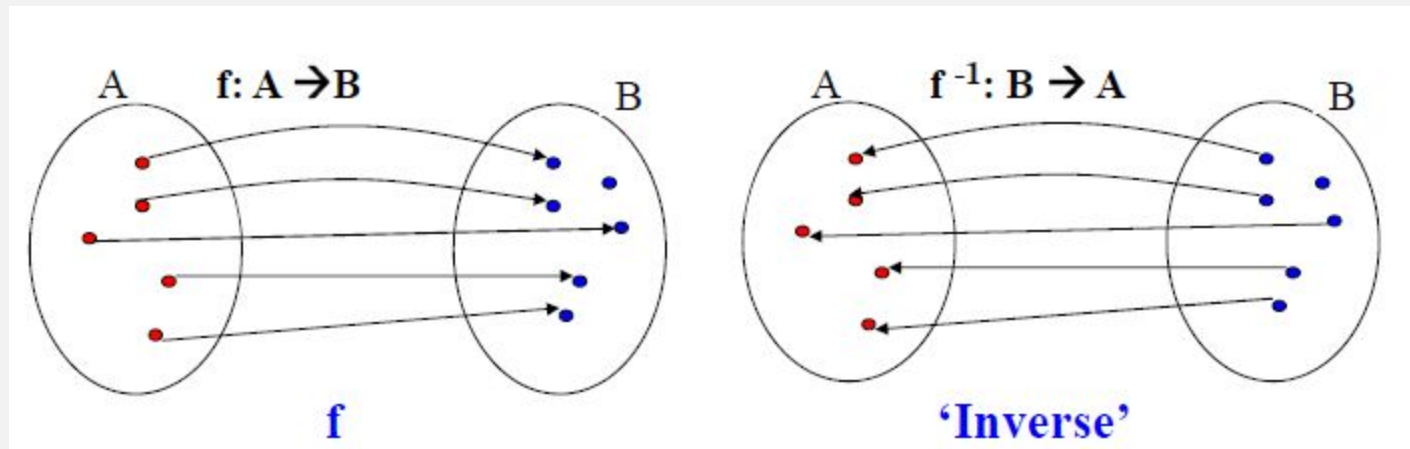


Inverse Function

Note: if f is not a bijection then it is not possible to define the inverse function of f . **Why?**

Assume f is **not onto**:

Inverse is **not a function**. One element of B is not assigned any value in A .



Inverse Function

Example 1:

- Let $A = \{1,2,3\}$ and i_A be the identity function
- $i_A(1) = 1$ $i_A^{-1}(1) = 1$
- $i_A(2) = 2$ $i_A^{-1}(2) = 2$
- $i_A(3) = 3$ $i_A^{-1}(3) = 3$
- Therefore, the inverse function of i_A is i_A .

Inverse Function

Example 2:

- Let $g : \mathbb{R} \rightarrow \mathbb{R}$, where $g(x) = 2x - 1$.
- What is the inverse function g^{-1} ?

Inverse Function

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- Let $g : \mathbb{R} \rightarrow \mathbb{R}$, where $g(x) = 2x - 1$.
- What is the inverse function g^{-1} ?

Approach to determine the inverse:

$$\begin{aligned} y &= 2x - 1 \Rightarrow y + 1 = 2x \\ &\Rightarrow (y+1)/2 = x \end{aligned}$$

- Define $g^{-1}(y) = x = (y+1)/2$

Test the correctness of inverse:

- $g(3) = ..$

Inverse Function

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- $g^{-1}(5) =$

Inverse Function

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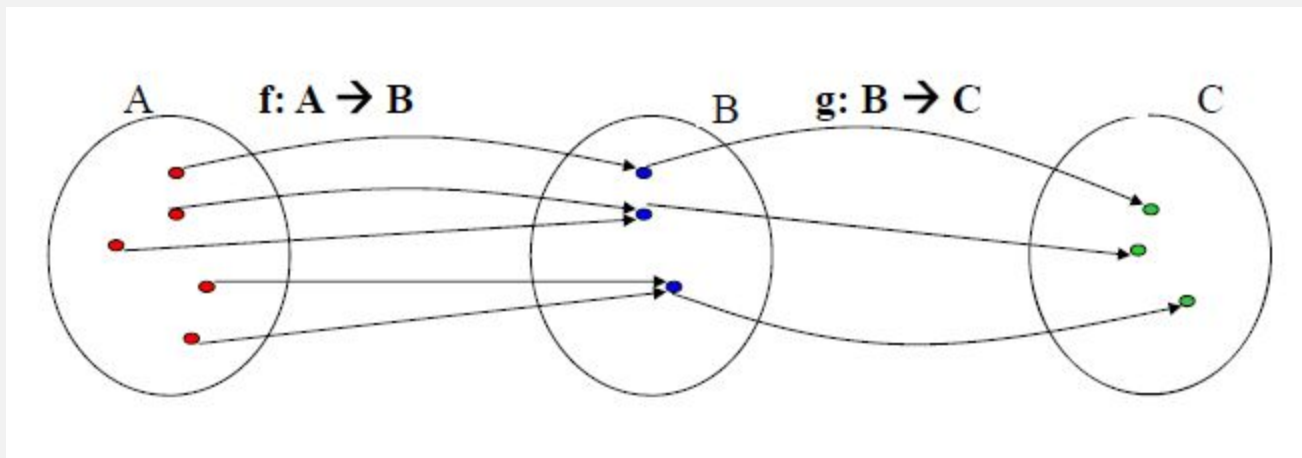
- Define $g^{-1}(y) = x = (y+1)/2$

Test the correctness of inverse:

- $g(3) = 2 \cdot 3 - 1 = 5$
- $g^{-1}(5) = (5+1)/2 = 3$

Composition of Function

Definition: Let f be a function from set A to set B and let g be a function from set B to set C . The composition of the functions g and f , denoted by $g \circ f$ is defined by $(g \circ f)(a) = g(f(a))$.



Composition of Function

Example 1:

- Let $A = \{1,2,3\}$ and $B = \{a,b,c,d\}$

$g : A \rightarrow A, \quad f : A \rightarrow B$

$1 \rightarrow 3 \quad 1 \rightarrow b$

$2 \rightarrow 1 \quad 2 \rightarrow a$

$3 \rightarrow 2 \quad 3 \rightarrow d$

$f \circ g : A \rightarrow B:$

- $1 \rightarrow ?$

Composition of Function

Example 1:

- Let $A = \{1,2,3\}$ and $B = \{a,b,c,d\}$

$g : A \rightarrow A, \quad f : A \rightarrow B$

$1 \rightarrow 3 \quad 1 \rightarrow b$

$2 \rightarrow 1 \quad 2 \rightarrow a$

$3 \rightarrow 2 \quad 3 \rightarrow d$

$f \circ g : A \rightarrow B:$

- $1 \rightarrow d$
- $2 \rightarrow b$
- $3 \rightarrow a$

Composition of Function

Example 2:

- Let f and g be two functions from Z to Z , where
- $f(x) = 2x$ and $g(x) = x^2$.
- $f \circ g : Z \rightarrow Z$
- $(f \circ g)(x) = f(g(x))$
 $= f(x^2)$
 $= 2(x^2)$
- $g \circ f : Z \rightarrow Z$
- $(g \circ f)(x) = ?$

Composition of Function

Example 2:

- Let f and g be two functions from Z to Z , where
- $f(x) = 2x$ and $g(x) = x^2$.
- $f \circ g : Z \rightarrow Z$
- $(f \circ g)(x) = f(g(x))$
 $= f(x^2)$
 $= 2(x^2)$
- $g \circ f : Z \rightarrow Z$
- $(g \circ f)(x) = g(f(x))$
 $= g(2x)$
 $= 4x^2$

Composition of Function

Example 3:

- $(f \circ f^{-1})(x) = x$ and $(f^{-1} \circ f)(x) = x$, for all x .
- Let $f : \mathbb{R} \rightarrow \mathbb{R}$, where $f(x) = 2x - 1$ and $f^{-1}(x) = (x+1)/2$.
- $$\begin{aligned}(f \circ f^{-1})(x) &= f(f^{-1}(x)) \\ &= f((x+1)/2) \\ &= 2((x+1)/2) - 1 \\ &= (x+1) - 1 \\ &= x\end{aligned}$$

Composition of Function

Example 3:

- $(f \circ f^{-1})(x) = x$ and $(f^{-1} \circ f)(x) = x$, for all x .
- Let $f : \mathbb{R} \rightarrow \mathbb{R}$, where $f(x) = 2x - 1$ and $f^{-1}(x) = (x+1)/2$.

- $$\begin{aligned}(f \circ f^{-1})(x) &= f(f^{-1}(x)) \\ &= f\left(\frac{x+1}{2}\right) \\ &= 2\left(\frac{x+1}{2}\right) - 1 \\ &= (x+1) - 1 \\ &= x\end{aligned}$$

- $$\begin{aligned}(f^{-1} \circ f)(x) &= f^{-1}(f(x)) \\ &= f^{-1}(2x - 1) \\ &= (2x)/2 \\ &= x\end{aligned}$$

Proofs

- **Definitions**: Describe the objects and notions that we use
- **Proofs**: is a logical argument that a statement is true
- **Theorem**: is a statement proved true
 - **Lemmas**: statements that are proved only because they assist the proofs of other statements (theorems/lemmas)
 - **Corollary**: statements related to the theorem that can be easily proved from the theorem or its proof

Types of Proofs

- Proof by Example ? Hand-waving?
- Proof by Construction
- Proof by Contradiction
- Proof by Induction

Proof by Construction (direct proof)

- Many theorems state that a particular object exists
 - One way to prove them is to show how to construct such objects
- In a constructive proof one attempts to demonstrate $p \rightarrow q$ directly
 - **Assume that p is true.**
 - **Use P to show that q must be true.**
- Example: if a and b are consecutive integers, then the sum $a+b$ is odd
 - Proof: as a and b are consecutive integers, $b = a+1$. Then $a+b = 2a+1$. That is, there exists an integer k such that $a+b = 2k+1$. Hence $a+b$ is odd
- Sometime we prove $p \rightarrow q$ by showing its contrapositive ($\neg q \rightarrow \neg p$) is true (contrapositive proof)

Proof by Contradiction

- The method of proof of $p \rightarrow q$ by contradiction.
 - 1. Assume that P is true.
 - 2. Assume that $\neg Q$ is true.
 - 3. Use P and $\neg Q$ to demonstrate a contradiction.
- Example: prove $\sqrt{2}$ is irrational
 - Proof: Assume $\sqrt{2}$ is rational, i.e., $\sqrt{2} = m/n$, m, n are integers (if m, n are divisible by same integer, divide them by it)
 - m, n both cannot be even
 - Now, $2n^2 = m^2$ and so m^2 is even
 - Therefore, m is also even
 - let $m = 2k$, for some integer k , then $2n^2 = 4k^2$
 - So, n^2 is even, hence n is also even.
 - Therefore m and n both are evens [Contradiction]

Proof by Induction

- Used to prove that all elements of some infinite set has a specific property
- For example,
 - To prove that $P(n)$ is true for all positive integers n , where $P(n)$ is a propositional function, we complete two steps:
 - **BASIS STEP:** We verify that $P(1)$ is true.
 - **INDUCTIVE STEP:** We show that the conditional statement $P(k) \rightarrow P(k + 1)$ is true for all positive integers k .
 - Expressed as a rule of inference
 - **$(P(1) \wedge \forall k(P(k) \rightarrow P(k + 1))) \rightarrow \forall nP(n)$, when the domain is the set of positive integers.**

Mathematical Induction

Example: Prove that $2^n < n!$ for all $n \geq 4$.

1. Basis Step: $P(4) = 2^4 < 4! = 16 < 24$ (trivial)

2. Inductive Step:

If $P(n)$ is true then $P(n+1)$ is also true

$$2^{n+1} = 2 \cdot 2^n$$

$$< 2 \cdot n! \quad (P(n) \text{ is true})$$

$$< (n+1)n! \quad (2 < n+1)$$

$$= (n+1)!$$

Strong Induction

- **The regular induction:**
 - basis step $P(1)$ and
 - inductive step $P(n-1) \rightarrow P(n)$
- **Strong induction uses:**
 - basis step $P(1)$ and
 - inductive step $P(1)$ **and** $P(2) \dots P(n-2)$ **and** $P(n-1) \rightarrow P(n)$
- **Example:** Show that a positive integer greater than 1 can be written as a product of primes.

Strong Induction

Example: Show that a positive integer greater than 1 can be written as a product of primes.

1. Basis step: $P(2)$ is true

2. Inductive step: Assume true for $P(2), P(3), \dots P(n)$

Show that $P(n+1)$ is true as well.

2 Cases:

- If $n+1$ is a prime then $P(n+1)$ is trivially true
- If $n+1$ is a composite then it can be written as a product of two integers $(n+1) = a \cdot b$ such that $1 < a, b < n+1$
- From the assumption $P(a)$ and $P(b)$ holds.
- Thus, $n+1$ can be written as a product of primes
- **End of proof**